

THIS IS AN EXTENDED VERSION OF THE PAPER:

Using the Median vs the Average with Randomized Quasi-Monte Carlo

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Abstract. Point sets for quasi-Monte Carlo (QMC) and randomized QMC (RQMC) integration are often constructed by selecting parameters to minimize a figure of merit (FOM) that measures the discrepancy between the empirical distribution of the points and the uniform distribution. Ideally, this FOM should give appropriate weights to the different subsets of coordinates to reflect their relative variance contributions. But these contributions (or sensitivity indices) are typically unknown and costly to estimate. A simpler alternative bypasses the FOMs and draw the parameters of the RQMC rule at random from some distribution. For popular QMC constructions, the probability of drawing poor parameters is very small. To take advantage of that, it was proposed recently to draw a small sample of QMC rules, estimate the integral with each of them, and take the median instead of the average as a final estimator. This way, the outliers (the rare bad rules) are unlikely to affect the estimator. To support this avenue, concentration inequalities for the median have been derived, and some achieve the best possible convergence rates in terms of the number of QMC points. We review these recent studies by putting them in a single general framework, we expand the set of known concentration inequalities for the median with RQMC, and we provide other new results and counter-examples. We also summarize numerical experiments that compare the mean square error (MSE) and the quantiles of the distribution for the average vs the median in this setting, for both QMC and RQMC.

This is the original extended version of a paper submitted to a journal, which had to be shortened to meet the page-limit constraints of the journal. The text and figures in dark red in this version do not appear in the shortened version. The numbers of the equations, figures, and references in this version may differ from those in the submitted version, because some of them were removed from the submitted version.

1 INTRODUCTION

We are interested in estimating an integral of the form

$$\mu = \mu(f) = \mathbb{E}[f(\mathbf{U})] = \int_{(0,1)^s} f(\mathbf{u}) d\mathbf{u}, \quad (1)$$

that represents the mathematical expectation of $f(\mathbf{U})$ with respect to a vector $\mathbf{U} = (U_1, \dots, U_s)$ of independent uniform random variables over the interval $(0, 1)$. With standard *Monte Carlo* (MC), these U_j are “simulated” by a uniform random number generator, and f represents all the transformations performed by the simulation program to produce the output $f(\mathbf{U})$ [1, 30, 43]. Here we assume that s is fixed to simplify the notation, although our discussion easily extends to the case where s is random and unbounded [30, 43]. Standard MC generates n independent uniform random points $\mathbf{U}_0, \dots, \mathbf{U}_{n-1}$, computes $f(\mathbf{U}_0), \dots, f(\mathbf{U}_{n-1})$, and estimates μ by the average

$$\hat{\mu}_{n,\text{mc}}(f) = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{U}_i). \quad (2)$$

If $\text{Var}[f(\mathbf{U})] < \infty$, which we assume for the rest of the paper, then $\text{Var}[\hat{\mu}_{n,\text{mc}}(f)] = O(n^{-1})$, so the width of a confidence interval on μ based on this MC estimator converges only at the slow rate of $O(n^{-1/2})$.

Under appropriate regularity conditions on f , *quasi-Monte Carlo* (QMC) and *randomized quasi-Monte Carlo* (RQMC) methods can improve this convergence rate [8, 11, 27, 43, 51]. QMC replaces the n independent random points U_i used by MC by a set of n deterministic points $P_n = \{u_0, \dots, u_{n-1}\}$ that cover the unit hypercube $[0, 1]^s$ more evenly than typical independent random points. It approximates μ by the deterministic average

$$\hat{\mu}_{n,\text{qmc}}(f) = \frac{1}{n} \sum_{i=0}^{n-1} f(u_i). \quad (3)$$

For certain Hilbert or Banach spaces $\mathcal{F}$ of functions f , one has an error bound of the form

$$e(f, P_n) := |\hat{\mu}_{n,\text{qmc}}(f) - \mu(f)| \leq \mathcal{D}_{\mathcal{F}}(P_n) \cdot \mathcal{V}_{\mathcal{F}}(f) \quad (4)$$

for all $f \in \mathcal{F}$, where $\mathcal{V}_{\mathcal{F}}(f)$ is a function of f only that measures its *variation* (it can be the norm of the space $\mathcal{F}$, but not always) and $\mathcal{D}_{\mathcal{F}}(P_n)$ is a function of P_n only that measures the *discrepancy* between the empirical distribution of P_n and the uniform distribution over $[0, 1]^s$. The definitions of both $\mathcal{V}_{\mathcal{F}}(f)$ and $\mathcal{D}_{\mathcal{F}}(P_n)$ depend on $\mathcal{F}$. In this setting, we define the worst-case integration error over $\mathcal{F}$ as

$$e(\mathcal{F}, P_n) := \sup_{f \in \mathcal{F}, \mathcal{V}_{\mathcal{F}}(f) \leq 1} e(f, P_n) \leq \mathcal{D}_{\mathcal{F}}(P_n). \quad (5)$$

For typical spaces $\mathcal{F}$, we know how to construct P_n such that $\mathcal{D}_{\mathcal{F}}(P_n) = O(n^{-\alpha}(\log n)^{s-1})$ where $\alpha > 1/2$ depends on $\mathcal{F}$. These constructions are often called *low discrepancy* point sets (some authors reserve this name for when $\alpha \geq 1$). When $\mathcal{V}_{\mathcal{F}}(f) < \infty$, $e(f, P_n)$ converges at the same rate as $\mathcal{D}_{\mathcal{F}}(P_n)$, which beats the MC rate when $\alpha > 1/2$.

With QMC, it is hard to estimate the variation $\mathcal{V}_{\mathcal{F}}(f)$ or the error $e(f, P_n)$ in practice. This is addressed by RQMC, which randomizes P_n into a random set $\tilde{P}_n = \{U_0, \dots, U_{n-1}\} \subset (0, 1)^s$ such that: (1) the uniformity of P_n as a whole is preserved; (2) each point U_i has the uniform distribution over $(0, 1)^s$ [28, 34, 35, 56]. A randomized point set that satisfies these two conditions is called an *RQMC point set*. It provides the *unbiased RQMC estimator*

$$\hat{\mu}_{n,\text{rqmc}}(f) = \frac{1}{n} \sum_{i=0}^{n-1} f(U_i), \quad (6)$$

with variance bound

$$\text{Var}[\hat{\mu}_{n,\text{rqmc}}(f)] = \mathbb{E}[e^2(f, \tilde{P}_n)] \leq \mathcal{D}_{\mathcal{F}}^2(P_n) \cdot \mathcal{V}_{\mathcal{F}}^2(f) \quad (7)$$

for a function space $\mathcal{F}$ and a discrepancy measure $\mathcal{D}_{\mathcal{F}}$ that are often the same as for QMC, but may also differ. It is common in RQMC settings to select the function space in a way that the discrepancy is invariant with respect to the randomization, so that randomizing the points does not change the discrepancy. In some cases, the convergence rate of $\text{MSE}[\hat{\mu}_{n,\text{rqmc}}(f)]$ may be even faster than that of $e^2(f, P_n)$; see [8, 11, 25, 56]. One would usually make r independent replicates of (6) for a small r (e.g., around 10 to 20) and estimate μ and $\text{Var}[\hat{\mu}_{n,\text{rqmc}}]$ by the average and sample variance of these r replicates. Since (6) itself is an average, this is a *sample average of averages*. Its variance will typically be

$$\text{Var}[\hat{\mu}_{n,\text{rqmc}}(f)]/r = O(\mathcal{D}_{\mathcal{F}}^2(P_n) \cdot \mathcal{V}_{\mathcal{F}}^2(f))/r = O(r^{-1}n^{-2\alpha+\epsilon})$$

for any $\epsilon > 0$. A central limit theorem (CLT) for the normal distribution does not always hold when $n \rightarrow \infty$ for r fixed [40], but a confidence interval on μ computed based on the r replicates is nevertheless often reasonably reliable [41]. Commonly-used randomizations for RQMC are discussed in Sections 3 and 4.

Important questions are how to choose the space $\mathcal{F}$ and the discrepancy measure $\mathcal{D}_{\mathcal{F}}(P_n)$, and how to construct P_n accordingly. A key observation is that when the dimension s is large, say more than 20 or so, filling up the hypercube

$(0, 1)^s$ uniformly is not practically feasible because it would take an excessive number of points. This explains the logarithmic term in the $O(n^{-\alpha}(\log n)^{s-1})$ discrepancy bound, which grows exponentially with s . It may suggest that QMC and RQMC cannot be effective when s is large, but this is not true. They often work well in high dimensions when f is close to a sum of low-dimensional functions, because then it suffices to have low integration error for each of these low-dimensional functions with the same point set P_n , or low variance for the same RQMC point set $\tilde{P}_n$, which is often possible [10, 27, 28, 43]. Specifically, one can apply the following ANOVA decomposition of f :

$$f(\mathbf{u}) = \sum_{\mathbf{u} \subseteq \{1, \dots, s\}} f_{\mathbf{u}}(\mathbf{u}) = \mu + \sum_{i=1}^s f_{\{i\}}(u_i) + \sum_{i,j=1}^s f_{\{i,j\}}(u_i, u_j) + \dots$$

in a way that the MC variance decomposes as

$$\text{Var}[f(\mathbf{U})] = \sigma^2 = \sum_{\mathbf{u} \subseteq \{1, \dots, s\}} \sigma_{\mathbf{u}}^2 = \sum_{\mathbf{u} \subseteq \{1, \dots, s\}} \text{Var}[f_{\mathbf{u}}(\mathbf{U})]$$

where $\mathbf{U}$ is a uniform vector in the unit hypercube in $|\mathbf{u}|$ dimensions. The quantities $\sigma_{\mathbf{u}}^2/\sigma^2$ for $\mathbf{u} \subseteq \{1, \dots, s\}$ are called the *Sobol' sensitivity indices* of the subsets of coordinates [67]. They represent the proportion of variance captured by each subset, and can be estimated by MC or RQMC; see [57, 68, 69] and the Online Appendix of [38]. Let

$$\hat{\mu}_{n,\text{rqmc}}(f_{\mathbf{u}}) = \frac{1}{n} \sum_{i=0}^{n-1} f_{\mathbf{u}}(\mathbf{U}_i) \quad (8)$$

denote the RQMC estimator of $\mathbb{E}[f_{\mathbf{u}}(\mathbf{U})]$. By applying (7) to $f_{\mathbf{u}}$, we find that $\text{Var}[\hat{\mu}_{n,\text{rqmc}}(f_{\mathbf{u}})] \leq \mathcal{D}_{\mathcal{F}}^2(P_n(\mathbf{u})) \cdot \mathcal{V}_{\mathcal{F}}^2(f_{\mathbf{u}})$ where $P_n(\mathbf{u})$ is the projection of P_n onto the subspace that contains the coordinates in $\mathbf{u}$. Under appropriate conditions that typically hold in RQMC settings, the RQMC variance also decomposes as above and we also obtain

$$\text{Var}[\hat{\mu}_{n,\text{rqmc}}(f)] = \sum_{\mathbf{u} \subseteq \{1, \dots, s\}} \text{Var}[\hat{\mu}_{n,\text{rqmc}}(f_{\mathbf{u}})] \leq \mathcal{D}_{\mathcal{F},\boldsymbol{\gamma}}^2(P_n) \cdot \mathcal{V}_{\mathcal{F},\boldsymbol{\gamma}}^2(f) \quad (9)$$

where $\boldsymbol{\gamma} = \{\gamma_{\mathbf{u}} > 0 : \mathbf{u} \subseteq \{1, \dots, s\}\}$ are weights on the subsets of coordinates,

$$\mathcal{D}_{\mathcal{F},\boldsymbol{\gamma}}^2(P_n) = \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}} \mathcal{D}_{\mathcal{F}}^2(P_n(\mathbf{u})) \quad \text{and} \quad \mathcal{V}_{\mathcal{F},\boldsymbol{\gamma}}^2(f) = \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^{-1} \mathcal{V}_{\mathcal{F}}^2(f_{\mathbf{u}}). \quad (10)$$

The role of the $\gamma_{\mathbf{u}}$'s is to give more weight to the important subsets of coordinates in the discrepancy measure $\mathcal{D}_{\mathcal{F},\boldsymbol{\gamma}}(P_n)$. With an appropriate function space for our target function f and a good estimate of appropriate weights, we can select P_n within a given type of parameterized constructions for which the parameters are chosen to minimize the weighted discrepancy $\mathcal{D}_{\mathcal{F},\boldsymbol{\gamma}}(P_n)$ [37]. By constructing the point sets in this way, the projections $P_n(\mathbf{u})$ of P_n that correspond to important subsets $\mathbf{u}$, which are the subsets having a significant sensitivity index $\sigma_{\mathbf{u}}^2/\sigma^2$, should have a good uniformity. This is typically achievable when all the important subsets are low-dimensional. In this setting, some of the parameters can be optimized while others could be fixed in advance or just chosen randomly, in which case they are part of the RQMC randomization. The estimation of appropriate weights, however, requires the estimation of the sensitivity indices and the identification of the important subsets, which can be very expensive computationally. Without knowledge of appropriate weights, there are simple heuristic solutions such as putting all the weights to 1 or putting weights $\gamma_{\mathbf{u}}$ that decrease geometrically with the cardinality of $\mathbf{u}$ at an arbitrary rate, but these heuristics do not always work well, because the weights may be inappropriate for the given function [16].

A different approach is to forget the idea of optimizing the parameters that define the point set and just draw them randomly and uniformly from an admissible set. Conditionally on the choice of parameters, the RQMC estimator is still

unbiased for μ , but its conditional variance may depend heavily on the parameters. It turns out that for popular RQMC constructions and function spaces $\mathcal{F}$, for a given $f \in \mathcal{F}$, most choices of parameters are very good in the sense that the conditional QMC error or RQMC variance given the choice of parameters is small, and only a very small proportion of the choices give a large (maybe huge) conditional error or variance. The unconditional distribution of the RQMC estimator is then likely to have high kurtosis, with most of the probability mass near the mean, but a tiny portion far away, giving occasional outliers that can be very far from the mean. One natural way to protect ourselves against these bad outliers is to replace the average of the r independent replicates of (6) by their median or by another robust mean estimator, as suggested and studied in [14, 16, 19, 58–60]. It was shown in these papers that in some QMC/RQMC situations, when certain parameters are selected at random, for $0 < \delta < 1/2$, the $(1 - \delta)$ -quantile of the QMC or RQMC error converges to zero with n faster for the median of r independent replicates than for their average, and essentially at the same rate as if we were selecting the parameters by minimizing the weighted discrepancy with the optimal weights.

The aim of this paper is to review, summarize, extend, and comment these results, discuss their practical implications, and examine more generally when the median would be better than the average, or vice-versa. Is the median better only in rare esoteric situations, or frequently? What is better depends only on the distribution of the RQMC estimator, so the answer depends on how this distribution behaves.

In Section 2, we provide general results on estimators of the expectation μ based on the median of independent replicates of a given estimator, and we discuss how these results are used to obtain concentration inequalities and bounds on the MSE of the median in general QMC and RQMC settings. In Sections 3 and 4, we examine in more details the convergence with the median vs the average for the two most popular types of QMC constructions, lattice rules of rank 1 and digital nets. For each, we discuss the choice of a discrepancy, error and variance bounds, which parameters can be optimized vs randomized, and what the distribution of the RQMC estimator looks like. In Section 5, we summarize our numerical experiments to compare various methods on a selection of examples. The GitHub site <https://github.com/pierrelecuyer/rqmc-mean-median> and the document [31] provide a much more extensive collection of results and plots, and links to the code used to make the experiments.

2 ESTIMATING AN EXPECTATION BY THE SAMPLE MEDIAN

In this section, we examine a robust approach that estimates the mean μ by the median of independent replicates of a given estimator. We provide basic results that are used in the following sections. Most of the discussion is made in a general setting; the estimators do not have to come from QMC or RQMC simulations. In the second subsection, we make connections with our QMC/RQMC setting. The types of techniques discussed here can also be used for Markov chain Monte Carlo (MCMC) and rare event simulation, for example [26, 52].

2.1 Concentration inequalities for the median

Let $X_1, \dots, X_r$ be r independent realizations of a random variable X that estimates an unknown quantity μ not necessarily equal to $\mathbb{E}[X]$. The most common way to combine $X_1, \dots, X_r$ into a single estimator of μ is to take the *sample average*

$$A_r = \frac{1}{r} \sum_{j=1}^r X_j. \quad (11)$$

This is fine when X has a friendly distribution concentrated around μ and with light tails, for example something that resembles a normal distribution. But when the distribution has a large kurtosis, outliers will cause occasional values of A_r that are very far from μ , so this estimator may become unreliable.

One way to improve the robustness to outliers is to replace the sample average by the *sample median*:

$$M_r = \text{med}(X_1, \dots, X_r) = X_{((r+1)/2)} \quad (12)$$

where $X_{(1)} \leq \dots \leq X_{(r)}$ are the *order statistics* and we assume for simplicity that r is always odd. When r is large enough, the chance that the median is an outlier becomes negligible, which means that the outliers have a negligible impact on the estimator M_r . This sketchy statement is made more accurate in the forthcoming Lemmas 2.1 to 2.3, which transform concentration inequalities for $|X - \mu|$ into inequalities for $|M_r - \mu|$. Suppose that

$$\mathbb{P}[|X - \mu| > \epsilon] \leq \delta \quad (13)$$

for some real numbers $\epsilon > 0$ and $0 < \delta \leq 1/2$. When we have a bound on the expected error $\mathbb{E}[|X - \mu|]$, an inequality of the form (13) can be obtained via the *Markov inequality*

$$\mathbb{P}[|X - \mu| > \epsilon] \leq \mathbb{E}[|X - \mu|] / \epsilon. \quad (14)$$

The following lemma refines the results given in Proposition 2.2 of [26] and Proposition 3.2 of [19]. Its proof makes a slight modification of the reasoning used in the proof of Proposition 2.1 of [52], which is re-used in [19, 26]. See [26, 52] for other relevant references.

LEMMA 2.1. *If r is odd and (13) holds for $\epsilon > 0$ and $0 < \delta \leq 1/2$, then*

$$\mathbb{P}[|M_r - \mu| > \epsilon] \leq 2^{r-1} \delta [\delta(1 - \delta)]^{(r-1)/2} = \delta [4\delta(1 - \delta)]^{(r-1)/2} =: \psi(r, \delta). \quad (15)$$

Proof. If the median M_r is not in the interval $[\mu - \epsilon, \mu + \epsilon]$, then the number Y of values of X_j among $X_1, \dots, X_r$ that fall outside that interval must be at least $(r + 1)/2$. This Y has a binomial(r, p) distribution with $p \leq \delta \leq 1/2$. Therefore

$$\begin{aligned} \mathbb{P}[|M_r - \mu| > \epsilon] &\leq \mathbb{P}[Y \geq (r + 1)/2] \leq \sum_{y=(r+1)/2}^r \binom{r}{y} \delta^y (1 - \delta)^{r-y} \\ &\leq \delta^{(r+1)/2} (1 - \delta)^{(r-1)/2} \sum_{y=(r+1)/2}^r \binom{r}{y} = 2^{r-1} \delta [\delta(1 - \delta)]^{(r-1)/2} = \delta [4\delta(1 - \delta)]^{(r-1)/2}. \quad \square \end{aligned}$$

For $\delta < 1/2$, the last term in (15) converges exponentially to zero as a function of r when $r \rightarrow \infty$, although that convergence is very slow when δ is close to $1/2$, and the bound is $1/2$ for all r when $\delta = 1/2$.

Remark 2.2. In [16], the rightmost term in (15) was $[4\delta]^{(r+1)/2}/4$ instead. If we take $r = 21$ and $\delta = 1/8$, for instance, we get $\delta [4\delta(1 - \delta)]^{(r-1)/2} = 3.211 \times 10^{-5}$ whereas $[4\delta]^{(r+1)/2}/4 = 1/2^{13} = 1.221 \times 10^{-4}$.

Lemma 2.1 may be sufficient for most practical purposes, but the following stronger result in which the “sup” is inside the probability also holds.

LEMMA 2.3. *Let $\{X(\xi), \xi \in \Xi\}$ be a family of random variables defined on the same probability space. Suppose there is a function $\mu : \Xi \rightarrow \mathbb{R}$ such that $D = \sup_{\xi \in \Xi} |X(\xi) - \mu(\xi)|$ is a well-defined random variable and $\mathbb{P}[D > \epsilon] \leq \delta$ for some constants $\epsilon > 0$ and $0 < \delta \leq 1/2$. Consider r independent realizations of $\{X(\xi), \xi \in \Xi\}$, giving $X_1(\xi), \dots, X_r(\xi)$ for all $\xi \in \Xi$. For each $\xi \in \Xi$, let $M_r(\xi) = \text{med}(X_1(\xi), \dots, X_r(\xi))$. Then,*

$$\mathbb{P} \left[\sup_{\xi \in \Xi} |M_r(\xi) - \mu(\xi)| > \epsilon \right] \leq \psi(r, \delta). \quad (16)$$

Proof. For any $\xi \in \Xi$, by applying Jensen's inequality to the median operator, we obtain

$$|M_r(\xi) - \mu(\xi)| = |\text{med}(X_1(\xi), \dots, X_r(\xi)) - \mu(\xi)| \leq \text{med}(|X_1(\xi) - \mu(\xi)|, \dots, |X_r(\xi) - \mu(\xi)|) \leq \text{med}(D_1, \dots, D_r)$$

where the $D_j = \sup_{\xi \in \Xi} |X_j(\xi) - \mu(\xi)|$ are independent and identically distributed. This gives $\sup_{\xi \in \Xi} |M_r(\xi) - \mu(\xi)| \leq \text{med}(D_1, \dots, D_r)$, and by applying Lemma 2.1 to $D_1, \dots, D_r$,

$$\mathbb{P} \left[\sup_{\xi \in \Xi} |M_r(\xi) - \mu(\xi)| > \epsilon \right] \leq \mathbb{P}[\text{med}(D_1, \dots, D_r) > \epsilon] \leq \psi(r, \delta). \quad \square$$

When (13) holds, we have the bound (17) on the p th moment of the absolute error $|M_r - \mu|$. Under the assumption that $|X - \mu|$ is bounded, we also have the bound (18), in which the last term decreases exponentially in r when $\delta_0 < 1/2$. For $p = 2$ these are bounds on the MSE of the median.

LEMMA 2.4. Let $\tau : (0, 1) \rightarrow (0, \infty)$ be a strictly decreasing differentiable function such that (13) holds with $\epsilon = \tau(\delta)$ for any $\delta \in (0, \delta_0)$ for some $\delta_0 \leq 1/2$, and let $\epsilon_0 = \tau(\delta_0)$. For all odd r and $p > 0$, we have

$$\mathbb{E}[|M_r - \mu|^p] \leq \epsilon_0^p + \int_{\epsilon_0}^{\infty} \psi(r, \tau^{-1}(\epsilon)) p \epsilon^{p-1} d\epsilon = \epsilon_0^p + \int_0^{\delta_0} \psi(r, \delta) p (\tau(\delta))^{p-1} |\tau'(\delta)| d\delta. \quad (17)$$

Moreover, if $|X - \mu| \leq C$ for all realizations of X , then we also have

$$\mathbb{E}[|M_r - \mu|^p] \leq \epsilon_0^p + \psi(r, \delta_0)(C^p - \epsilon_0^p). \quad (18)$$

Proof. Using Lemma 2.1, and with the change of variable $\xi = \epsilon^p$, we have

$$\begin{aligned} \mathbb{E}[|M_r - \mu|^p] &= \int_0^{\infty} \mathbb{P}[|M_r - \mu|^p > \xi] d\xi = \int_0^{\infty} \mathbb{P}[|M_r - \mu| > \epsilon^p] p \epsilon^{p-1} d\epsilon \\ &\leq \int_0^{\epsilon_0} p \epsilon^{p-1} d\epsilon + \int_{\epsilon_0}^{\infty} \psi(r, \tau^{-1}(\epsilon)) p \epsilon^{p-1} d\epsilon \leq \epsilon_0^p + \int_{\epsilon_0}^{\infty} \psi(r, \tau^{-1}(\epsilon)) p \epsilon^{p-1} d\epsilon. \end{aligned}$$

The last equality in (17) comes from change of variable $\epsilon = \tau(\delta)$. For (18), in case $|X - \mu| \leq C$ for all X , the integral with respect to ϵ can stop at C instead of ∞ , and then the first integral in (17) becomes

$$\int_{\epsilon_0}^C \psi(r, \tau^{-1}(\epsilon)) p \epsilon^{p-1} d\epsilon \leq \psi(r, \tau^{-1}(\epsilon_0)) \int_{\epsilon_0}^C p \epsilon^{p-1} d\epsilon = \psi(r, \delta_0)(C^p - \epsilon_0^p). \quad \square$$

COROLLARY 2.5. Under the conditions of Lemma 2.4, for the special case in which $\tau(\delta) = K\delta^{-\beta}$ for some positive constants K and β , if $q := (r-1)/2 - p\beta \geq 1$ is large enough and $\delta_0 < 1/2$, we have

$$\mathbb{E}[|M_r - \mu|^p] \leq K^p \left[\delta_0^{-p\beta} + 4^{p\beta-1} p\beta \frac{(\delta_0(1-\delta_0))^{q+1}}{q(1-2\delta_0)} \right]. \quad (19)$$

For $\delta_0 < 1/2$, we have $\rho_0 := 4\delta_0(1-\delta_0) < 1$, so the last term in the brackets decreases exponentially as $O(\rho_0^q/q)$, but the first one is always larger than $2^{p\beta}$.

Proof. With $\tau(\delta) = K\delta^{-\beta}$, we have $\epsilon_0 = K\delta_0^{-\beta}$, $\tau'(\delta) = -K\beta\delta^{-\beta-1}$, and $\psi(r, \delta) = \delta(4\delta(1-\delta))^{(r-1)/2}$ for $\delta \leq 1/2$. By putting this in (17) and integrating, and using the incomplete beta function

$$B_{\delta_0}(q+1, q+1) = \int_0^{\delta_0} (\delta(1-\delta))^q d\delta \approx \frac{(\delta_0(1-\delta_0))^{q+1}}{q(1-2\delta_0)}$$

when q is large, we obtain

$$\begin{aligned} \mathbb{E}[|M_r - \mu|^p] &\leq \epsilon_0^p + pK^p\beta \int_0^{\delta_0} (4\delta(1-\delta))^{(r-1)/2} \delta^{-p\beta} d\delta \leq (K\delta_0^{-\beta})^p + 2^{r-1}pK^p\beta B_{\delta_0}(q+1, q+1) \\ &\approx (K\delta_0^{-\beta})^p + 2^{r-1}pK^p\beta \frac{(\delta_0(1-\delta_0))^{q+1}}{q(1-2\delta_0)} = K^p \left[\delta_0^{-p\beta} + 4^{p\beta-1}p\beta \frac{(4\delta_0(1-\delta_0))^{q+1}}{q(1-2\delta_0)} \right]. \quad \square \end{aligned}$$

We will use this corollary to bound the moments of the median by using explicit expressions for $\tau(\delta)$ available for special cases, where $K = K(n)$ is a function of n which typically decreases as $O(n^{-\beta})$. Then, the bound (19) converges to 0 when $n \rightarrow \infty$, but not when n is fixed and $r \rightarrow \infty$, because $\delta_0^{-p\beta} > 2^{p\beta}$ does not decrease with r . For this same reason, increasing r together with n will not give a better convergence rate for the bound than increasing only n while keeping r fixed. These statements are for the bound. It may occur that $\mathbb{E}[|M_r - \mu|^p]$ itself converges faster than the bound. For fixed n , if X has a nonzero density in the neighborhood of the median, then M_r will converge to the true median with probability 1 when $r \rightarrow \infty$, and it also obeys a central limit theorem, by standard theory on quantile estimation [63]. It will converge to μ if and only if the true median is equal to μ . The latter happens in particular in the special case where X has a symmetric distribution, but often does not hold. Sometimes, X has no density or probability mass in the neighborhood of the median, in which case M_r will not converge at all when $r \rightarrow \infty$.

2.2 Application to QMC and RQMC

In this paper, in the QMC settings, the point set $P_n = P_{n,\theta}$ depends on a vector θ of parameters drawn with a given distribution (typically uniform) from a discrete set. Then, $X = X(f) = \widehat{\mu}_{n,\text{qmc}}(f)$ is a random variable and the probabilities and expectations in the forthcoming equations for QMC are with respect to the distribution of θ . For any $f \in \mathcal{F}$, the function space, and given realization of θ , the error bound (4) can be written as

$$|X(f) - \mu(f)| = |\widehat{\mu}_{n,\text{qmc}}(f) - \mu(f)| \leq \mathcal{D}_{\mathcal{F}}(P_{n,\theta}) \cdot \mathcal{V}_{\mathcal{F}}(f)$$

and therefore $e(\mathcal{F}, P_{n,\theta}) \leq \mathcal{D}_{\mathcal{F}}(P_{n,\theta})$. A common way to prove the existence of point sets $P_{n,\theta}$ for which $e(\mathcal{F}, P_{n,\theta})$ decreases at a certain rate as a function of n is to first show that the average value of the discrepancy to the power 2λ , over all admissible choices of θ , satisfies

$$\mathbb{E}[\mathcal{D}_{\mathcal{F}}^{2\lambda}(P_{n,\theta})] \leq b(n, \lambda) < \infty \quad (20)$$

for all $\lambda \in (\lambda_{\min}, 1]$, where the bound $b(n, \lambda)$ typically has the form $b(n, \lambda) = c_s(\lambda)n^{-\alpha+\epsilon'}$ for some $\alpha \geq 1$, where $\epsilon' > 0$ is an increasing function of λ that can be made arbitrarily small by taking λ very close to $\lambda_{\min}$, and $c_s(\lambda)$ is a constant that may depend on the dimension s and on λ but not on n , and increases towards infinity when $\lambda \searrow \lambda_{\min}$. If the average satisfies the bound in (20), then there must be at least one choice of θ for which $\mathcal{D}_{\mathcal{F}}(P_{n,\theta}) \leq (b(n, \lambda))^{1/2\lambda}$. This argument is used for example in [8, 11, 12, 64] and other references given there, where 2λ is sometimes denoted by $1/2\tau$. The reason for taking the power 2λ or $1/2\tau$ instead of λ or $1/\tau$ is that $\mathcal{D}_{\mathcal{F}}^2(P_{n,\theta})$ can usually be expressed nicely as a sum (see Section 3.1, for example). By taking $\lambda = 1$, the bound in (20) is usually no better than $O(1/n)$, because the expected discrepancy does not converge faster than that, due to a small proportion of very large values. Taking $\lambda < 1$ reduces the impact of these large values, and this is typically required to get $\alpha > 1$ [19, Section 3.1]. When this happens and we take r random choices of θ , the median is likely to converge at a better rate than the average.

We now develop concentration inequalities for X and for the median. When (20) holds, Markov's inequality yields

$$\mathbb{P} \left[\sup_{f \in \mathcal{F}, \mathcal{V}(f) \leq 1} |X(f) - \mu(f)| > (b(n, \lambda)/\delta)^{1/2\lambda} \right] = \mathbb{P} \left[\mathcal{D}_{\mathcal{F}}^{2\lambda}(P_{n, \theta}) > b(n, \lambda)/\delta \right] \leq \delta \quad (21)$$

for any $\delta > 0$. For $0 < \delta < 1/2$, applying Lemma 2.1 then gives

$$\sup_{f \in \mathcal{F}, \mathcal{V}(f) \leq 1} \mathbb{P} \left[|M_r(f) - \mu(f)| > (b(n, \lambda)/\delta)^{1/2\lambda} \right] \leq \psi(r, \delta), \quad (22)$$

and using Lemma 2.3 together with (21) yields the stronger result:

$$\mathbb{P} \left[\sup_{f \in \mathcal{F}, \mathcal{V}(f) \leq 1} |M_r(f) - \mu(f)| > (b(n, \lambda)/\delta)^{1/2\lambda} \right] \leq \psi(r, \delta). \quad (23)$$

For the RQMC setting, $P_n = P_{n, \theta}$ may still depend on a vector θ of parameters drawn at random before applying the additional RQMC randomization that gives $\tilde{P}_n$ and makes the estimator $X = \hat{\mu}_{n, \text{rqmc}}(f)$ unbiased for μ . There are then two levels of randomness, and in the following we will add a “tilde” over the $\mathbb{E}$ or $\mathbb{P}$ when the expectation or the probability is with respect to both θ and the RQMC randomization. We can make a similar reasoning as for QMC, but with (20) replaced by $\tilde{\mathbb{E}} \left[\mathcal{D}_{\mathcal{F}}^{2\lambda}(\tilde{P}_{n, \theta}) \right] \leq b(n, \lambda) < \infty$. If this and (4) hold, then

$$\tilde{\mathbb{E}} \left[\sup_{f \in \mathcal{F}, \mathcal{V}(f) \leq 1} |X(f) - \mu(f)|^{2\lambda} \right] \leq \tilde{\mathbb{E}} \left[\mathcal{D}_{\mathcal{F}}^{2\lambda}(\tilde{P}_{n, \theta}) \right] \leq b(n, \lambda) < \infty, \quad (24)$$

which implies by Markov inequality that

$$\tilde{\mathbb{P}} \left[\sup_{f \in \mathcal{F}, \mathcal{V}(f) \leq 1} |X(f) - \mu(f)| > (b(n, \lambda)/\delta)^{1/2\lambda} \right] \leq \delta, \quad (25)$$

and therefore for $0 < \delta < 1/2$, for the median of r RQMC estimators, we obtain from Lemma 2.3 that

$$\tilde{\mathbb{P}} \left[\sup_{f \in \mathcal{F}, \mathcal{V}(f) \leq 1} |M_r(f) - \mu(f)| > (b(n, \lambda)/\delta)^{1/2\lambda} \right] \leq \psi(r, \delta). \quad (26)$$

We can apply Lemma 2.4 and Corollary 2.5 to the above settings by taking $\epsilon = \tau(\delta) = K\delta^{-\beta} = (b(n, \lambda)/\delta)^{1/2\lambda}$. That is, $\beta = 1/2\lambda$ and $K = K(n) = (b(n, \lambda))^{1/2\lambda}$.

2.3 Some insight on when the median could beat the average

When is it better to estimate the mean μ by the median M_r rather than the average A_r ? The following simplified examples provide insight on this by pointing out situations where the average is much better than the median, and situations where we have the opposite.

Example 2.6. Suppose X has a discrete distribution on $\{-a, 0, b\}$, with $\mathbb{P}[X = 0] = p_0$, $\mathbb{P}[X = -a] = p_a$, $\mathbb{P}[X = b] = p_b$. The exact mean is $\mu = \mathbb{E}[X] = bp_b - ap_a$. If $p_0 = 0$, then the empirical median is always either a or b , i.e., one of the two extreme points of the distribution, usually far from the true mean, so the median is a very bad estimator.

For the symmetric case where $a = b$ and $p_a = p_b = p = (1 - p_0)/2$, we have $\mu = 0$, $\text{Var}[X] = \mathbb{E}[X^2] = pa^2$, $\mathbb{E}[X^4] = pa^4$, and the kurtosis of X is $\kappa = \mathbb{E}[X^4]/\mathbb{E}^2[X^2] = 1/p$. We also have $\text{MSE}[A_r] = \text{Var}[X]/r = pa^2/r$ and

$$\text{MSE}[M_r] = \mathbb{E}[M_r^2] = a^2\mathbb{P}[|M_r| = a] = 2a^2\mathbb{P}[B \geq (r+1)/2] \leq 2a^2\psi(r, p) \leq a^22^r p^{(r+1)/2},$$

where B is a binomial(r, p) random variable. This holds because $|M_r| = a$ if and only if at least $(r+1)/2$ of the X_j are a or at least $(r+1)/2$ of the X_j are $-a$. Then the MSE ratio between the median and the average satisfies

$$\text{MSE}[M_r]/\text{MSE}[A_r] = 2r\mathbb{P}[B \geq (r+1)/2]/p \leq 2r(4p)^{(r-1)/2}. \quad (27)$$

The ratio does not depend on a and the upper bound in (27) is less than 1 if and only if $p < (2r)^{-2/(r-1)}/4$, which is an increasing function of r . For example, for $r = 3$ we need $p < 0.0417$, whereas for $r = 41$ we need $p < 0.2006$. Note that a small p corresponds to a large kurtosis. If $p = 1/2$, an extreme case, then $p_0 = 0$ and the empirical median will always be either $-a$ or a , which is never better than the average. But when p_0 and r are large, the median M_r will be 0 (the exact mean) most of the time, and therefore better than the average.

For an asymmetric example, suppose $p_a = p_b = p$ is small, but $b \gg a$. Then the median will still be 0 most of the time, while the exact mean is much larger than 0. In this skewed case, the average does much better than the median, even if the kurtosis can be large. A similar situation occurs if $a = b$ and $p_a \ll p_b < 1/4$. In general, the median rarely does well for skewed distributions of X , because it introduces bias by “clipping out” the tails in some sense.

This stylized example indicates that when X has a (more general) bimodal distribution, the empirical median is usually a bad estimator of the mean. For a trimodal distribution, if the distribution is symmetric and the probability mass of the central mode is large enough, the empirical median will perform well. But if the central mode has a small weight or the distribution is highly asymmetric, the median is likely to perform more poorly than the average. Also, if X has a highly skewed distribution, its true median will be far from its mean, in which case the empirical median is likely to be a poor estimator of the mean. In a nutshell, which estimator is more appropriate depends very much on the distribution of X , and there are extreme bad cases for both estimators.

Example 2.7. Suppose X has a symmetric continuous distribution centered at μ , with density $g(x) > 0$ in a neighborhood of μ , and $\text{Var}[X] = \sigma^2 < \infty$. Then the median and the mean are μ , and both A_r and M_r are unbiased estimators of μ , so their MSE is their variance. We know that $\text{Var}[A_r] = \sigma^2/r$ and also that $2\sqrt{r}(M_r - \mu)g(\mu)$ converges to a standard normal distribution when $r \rightarrow \infty$ [30, Theorem 5.8 (ii)] or [63]. This means that for large r , $\text{Var}[M_r] \approx 1/[4rg^2(\mu)]$, which beats $\text{Var}[A_r]$ if and only if $g(\mu)\sigma > 1/2$. That is, the median wins asymptotically when $g(\mu)\sigma$ is large enough. As a special case, if X has a normal distribution, then $g(\mu)\sigma = 1/\sqrt{2\pi} < 1/2$, and the ratio $\text{Var}[M_r]/\text{Var}[A_r]$ is 1 for $r = 1$, ≈ 1.346 for $r = 3$, and increases towards $\pi/2 \approx 1.5708$ when $r \rightarrow \infty$.

The last example shows that when X has a distribution that is not far from normal, A_r is usually a better estimator than M_r . This would occur, for example, if X is an RQMC estimator $\hat{\mu}_{n,\text{rqmc}}(f)$ that obeys a CLT towards the normal distribution when $n \rightarrow \infty$. This happens in particular for a digital net randomized by the nested uniform scramble (NUS) of Owen [45], as pointed out by [18] and others, but it does not occur for a randomly-shifted lattice rule [40]. Highly non-normal distributions of X , e.g., with large kurtosis and sometimes skewed, do occur in RQMC, as shown in [41] and in Section 5. In our experiments, we have observed values of the inverse ratio $\text{MSE}[A_r]/\text{MSE}[M_r]$ up to around 10^{20} . See Section 5.3 in the extended version. If X is an unbiased estimator with an asymmetric distribution and the median differs from the mean, then $\text{MSE}[M_r]/\text{MSE}[A_r] \rightarrow \infty$ when $r \rightarrow \infty$.

2.4 Using the median with a correction

Using the median to estimate the mean usually gives a biased estimator, and the bias can be important when the distribution is highly asymmetric. Several ways of modifying the median have been proposed to try to reduce the overall

MSE. See for example [4, 14] and the references given there. We tried some of them in our experiments, including the Lee-Valiant (LV) estimator from [42], as given in Definition 3.4 of [14], and none gave us a significant improvement.

3 LATTICE RULES

3.1 Rank-1 lattice rules in a weighted Korobov space of functions

A *lattice rule of rank 1* in s dimensions is a QMC integration method defined by selecting a number of points n and a generating vector $\mathbf{a} = (a_1, \dots, a_s) \in \{1, \dots, n-1\}^s$. The QMC points are $\mathbf{u}_i = (i/n)\mathbf{a} \bmod 1$ for $i = 0, \dots, n-1$. We also impose that $\gcd(a_j, n) = 1$ for all j to make sure that all projections $P_n(\mathbf{u})$ of P_n have n distinct points. The standard way of randomizing these points for RQMC is a *random shift modulo 1* (RS): generate a single random point $\mathbf{V}$ uniformly in $(0, 1)^s$ and add it to each $\mathbf{u}_i$, modulo 1 [2]. We write the shifted points as $\tilde{\mathbf{u}}_i = \mathbf{u}_i \oplus_1 \mathbf{V} := \mathbf{u}_i + \mathbf{V} \bmod 1$, and the resulting shifted point set as $\tilde{P}_n = P_n \oplus_1 \mathbf{V}$.

Lattice rules are designed primarily for one-periodic smooth functions f with Fourier expansion

$$f(\mathbf{u}) = \sum_{\mathbf{h} \in \mathbb{Z}^s} \hat{f}(\mathbf{h}) e^{2\pi i \mathbf{h}^\top \mathbf{u}} \quad \text{for } \mathbf{u} \in [0, 1]^s,$$

with Fourier coefficients $\hat{f}(\mathbf{h}) = \int_{(0,1)^s} f(\mathbf{u}) \exp(-2\pi i \mathbf{h}^\top \mathbf{u}) d\mathbf{u}$. For a lattice rule applied to f , it is known that

$$e(f, P_n) = \sum_{\mathbf{0} \neq \mathbf{h} \in L_s^*} \hat{f}(\mathbf{h})$$

if the Fourier expansion is absolutely convergent, and for a randomly-shifted lattice rule, we have

$$\text{Var}[\hat{\mu}_{n,\text{rqmc}}(f)] = \sum_{\mathbf{0} \neq \mathbf{h} \in L_s^*} |\hat{f}(\mathbf{h})|^2,$$

where $L_s^* = \{\mathbf{h} \in \mathbb{R}^s : \mathbf{h}^\top \mathbf{v} \in \mathbb{Z} \text{ for all } \mathbf{v} \in P_n\} \subseteq \mathbb{Z}^s$ is the dual of the lattice generated by P_n [34, 65].

For an integer $\alpha > 0$, we consider the class $\mathcal{F}_\alpha$ of functions f with square-integrable mixed partial derivatives up to order α , and for which the periodic continuation of its derivatives up to order $\alpha - 1$ is continuous across the unit cube boundaries (i.e., continuous over the unit torus). This is a Sobolev space restricted to one-periodic functions. It is equivalent to a reproducing kernel Hilbert space with a Korobov kernel, which is more general since it also allows non-integer values of α [8], but in practice α is almost always taken as an integer. The periodic continuity conditions may look too strong, but a continuous function over $[0, 1]^s$ can easily be made continuous and periodic over the s -dimensional unit torus by applying a baker (or tent) transformation after the RS [22]. This transformation replaces each $u_{i,j}$ by $\text{baker}(u_{i,j}) = 1 - 2|u_{i,j} - 1/2|$. See also [8, Chapter 7] for other ways to handle this.

We now introduce weights $\gamma_{\mathbf{u}}$ as in (10). As in [8, 12, 39, 64], we consider a weighted version of the space $\mathcal{F}_\alpha$ in which the square variation is

$$\mathcal{V}_{\gamma,\alpha}^2(f) = \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \frac{1}{\gamma_{\mathbf{u}}} \frac{1}{(4\pi^2)^{\alpha|\mathbf{u}|}} \int_{[0,1]^{|\mathbf{u}|}} \left| \frac{\partial^{\alpha|\mathbf{u}|}}{\partial \mathbf{u}^\alpha} f_{\mathbf{u}}(\mathbf{u}) \right|^2 d\mathbf{u} \quad (28)$$

and the corresponding square discrepancy is

$$\mathcal{P}_{2\alpha,\gamma}(P_n) = \sum_{\mathbf{0} \neq \mathbf{h} \in L_s^*} \gamma_{\mathbf{u}(\mathbf{h})} \prod_{j \in \mathbf{u}(\mathbf{h})} |h_j|^{-2\alpha}, \quad (29)$$

where $\mathbf{u}(\mathbf{h}) = \mathbf{u}(h_1, \dots, h_s)$ is the set of indices j for which $h_j \neq 0$. We denote by $\mathcal{F}_{\gamma,\alpha}$ the space of functions $f \in \mathcal{F}_\alpha$ for which $\mathcal{V}_{\gamma,\alpha}(f) < \infty$ and by $\mathcal{F}_{\gamma,\alpha}^{(1)}$ the subset of those for which $\mathcal{V}_{\gamma,\alpha}(f) \leq 1$. When α is a positive integer, $\mathcal{P}_{2\alpha,\gamma}(P_n)$

can be computed by the finite sum

$$\mathcal{P}_{2\alpha, \gamma}(P_n) = \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}} \left[\frac{(-4\pi^2)^\alpha}{(2\alpha)!} \right]^{|\mathbf{u}|} \frac{1}{n} \sum_{i=0}^{n-1} \prod_{j \in \mathbf{u}} B_{2\alpha}(u_{i,j}),$$

where $(u_{i,1}, \dots, u_{i,s}) = \mathbf{u}_i = \mathbf{ia}/n \bmod 1$ and $B_{2\alpha}$ is the Bernoulli polynomial of order 2α . See [8, 10, 12, 21, 27, 38, 39, 64] for further details. For QMC with a deterministic lattice rule (no RS) and $f \in \mathcal{F}_{\gamma, \alpha}$, we have the error bound

$$e(f, P_n) = |\widehat{\mu}_{n, \text{qmc}}(f) - \mu(f)| \leq \mathcal{V}_{\gamma, \alpha}(f) \cdot (\mathcal{P}_{2\alpha, \gamma}(P_n))^{1/2}, \quad (30)$$

while for RQMC with a randomly-shifted lattice rule, we have the variance bound

$$\text{Var}[\widehat{\mu}_{n, \text{rqmc}}(f)] \leq \mathcal{V}_{\gamma, \alpha}^2(f) \cdot \mathcal{P}_{2\alpha, \gamma}(P_n). \quad (31)$$

Therefore,

$$e^2(\mathcal{F}_{\gamma, \alpha}, P_n) = \sup_{f \in \mathcal{F}_{\gamma, \alpha}^{(1)}} \text{Var}[\widehat{\mu}_{n, \text{rqmc}}(f)] = \mathcal{P}_{2\alpha, \gamma}(P_n). \quad (32)$$

It is known how to construct rank-1 lattices (a vector $\mathbf{a} = \mathbf{a}(n)$ for each n) such that $\mathcal{P}_{2\alpha, \gamma}(P_n) = O(n^{-2\alpha+\epsilon})$ for any $\epsilon > 0$, for any given vector of weights $\gamma_{\mathbf{u}}$ [8, 12, 37, 53, 64]. But finding appropriate weights $\gamma_{\mathbf{u}}$ is generally not easy. By exploiting the fact that $\mathcal{P}_{2\alpha, \gamma}(P_n)$ is the RQMC variance (and squared QMC error) for the (known) worst-case function among those with $\mathcal{V}_{\gamma, \alpha}^2(f) \leq 1$, [38] point out that for this worst-case function, $\sigma_{\mathbf{u}}^2 = \gamma_{\mathbf{u}} [|B_{2\alpha}(0)| (4\pi^2)^\alpha / (2\alpha)!]^{|\mathbf{u}|}$. If estimates of the $\sigma_{\mathbf{u}}^2$'s are available, one way to select the weights is to pretend that our function is the worst-case function and use the corresponding weights $\gamma_{\mathbf{u}}$. For example, for $\alpha = 1$ this gives $\gamma_{\mathbf{u}} = (3/\pi^2)^{|\mathbf{u}|} \sigma_{\mathbf{u}}^2$. Estimating the variance components $\sigma_{\mathbf{u}}^2$ is doable [61] but can be costly in practice.

3.2 The median estimator for lattice rules with a random generating vector

An alternative approach that requires no information on the weights is to simply pick the generating vector $\mathbf{a}$ at random uniformly from $\mathcal{A}_n = \{\mathbf{a} : 1 \leq a_j < n \text{ and } \gcd(a_j, n) = 1 \text{ for all } j\}$, independently across RQMC replications. With this approach, some bad choices of $\mathbf{a}$ give a large QMC error or a large conditional RQMC variance, which results in a large MSE for the unconditional QMC estimator (3) or RQMC estimator (6) with random $\mathbf{a}$; see Example 3.5. Fortunately, only a tiny proportion of the vectors $\mathbf{a} \in \mathcal{A}_n$ are bad. By taking the median instead of the average of the QMC/RQMC replicates, we can remove their impact with very high probability and obtain a high-quality estimator.

Goda and L'Ecuyer [16] studied the *median* estimator M_r for QMC with a lattice rule in the weighted Korobov space considered above (they use $\gamma_{\mathbf{u}}^2$ in place of our more standard $\gamma_{\mathbf{u}}$; this explains some discrepancies between their expressions and ours). They proved the following Theorem 3.1 (in their Theorem 2.5 and Corollary 2.6). Let $\varphi(n)$ be the number of positive integers smaller than n that are relatively prime to n (Euler's totient function). We recall that for prime n , we have $\varphi(n) = n - 1$, and for any $n \geq 3$, $n/\varphi(n) \leq 1.781073 \log \log n + 2.50637/\log \log n = O(n^{\epsilon'})$ for all $\epsilon' > 0$. Note that the expression $1/\varphi(n)^{\alpha-\epsilon'}$ inside the probabilities in (33) and (37) can decrease almost as $O(\varphi(n)^{-\alpha})$, since ϵ' can be taken arbitrarily small, and therefore as $O(n^{-\alpha+\epsilon''})$ for any $\epsilon'' > 0$.

THEOREM 3.1. *Let $n \geq 2$ be an integer, let $\mathbf{a}$ be a vector drawn uniformly from $\mathcal{A}_n$, and let $P_n = P_{n, \theta}$ be the rank-1 lattice point set generated by $\theta = \mathbf{a}$. For any smoothness $\alpha > 1/2$, $0 < \epsilon' < \alpha$, $0 < \delta < 1/2$, and choice of weights γ , there is a constant c_1 such that*

$$\mathbb{P} \left[e(\mathcal{F}_{\gamma, \alpha}, P_{n, \theta}) > \frac{c_1}{(\delta \varphi(n))^{\alpha-\epsilon'}} \right] \leq \delta. \quad (33)$$

PROOF. We use the general results of Section 2 to write a more compact proof than in [16]. Recall that $e^2(\mathcal{F}_{\gamma,\alpha}, P_{n,\theta}) = \mathcal{P}_{2\alpha,\gamma}(P_{n,\theta})$. The authors of [16] prove that the *average* value of $(\mathcal{P}_{2\alpha,\gamma}(P_n))^\lambda$ over all admissible choices of the generating vector, for any $\lambda \in (1/2\alpha, 1]$, satisfies

$$\mathbb{E}[e((\mathcal{F}_{\gamma,\alpha}, P_{n,\theta}))^{2\lambda}] = \mathbb{E}[(\mathcal{P}_{2\alpha,\gamma}(P_{n,\theta}))^\lambda] \leq c_{s,\alpha}(\lambda)/\varphi(n) \quad (34)$$

where

$$c_{s,\alpha}(\lambda) = \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^\lambda (2\zeta(2\alpha\lambda))^{|\mathbf{u}|}, \quad (35)$$

in which ζ is the Riemann zeta function, $\zeta(x) = \sum_{i=1}^{\infty} i^{-x}$ for $x > 1$. Then for any $\delta > 0$, Markov's inequality yields

$$\mathbb{P}\left[e(\mathcal{F}_{\gamma,\alpha}, P_{n,\theta}) > \left(\frac{c_{s,\alpha}(\lambda)}{\delta\varphi(n)}\right)^{1/2\lambda}\right] = \mathbb{P}\left[(\mathcal{P}_{2\alpha,\gamma}(P_{n,\theta}))^\lambda > \frac{c_{s,\alpha}(\lambda)}{\delta\varphi(n)}\right] \leq \delta. \quad (36)$$

We want the exponent $1/2\lambda$ on the left to be as large as possible to make that bound as small as possible, but one must have $2\alpha\lambda > 1$ for the constant $c_{s,\alpha}(\lambda)$ to be finite, so we can take λ just slightly larger than $1/2\alpha$. By taking $\lambda = 1/2(\alpha - \epsilon')$ and then $c_1 = [c_{s,\alpha}(\lambda)]^{1/2\lambda}$, we have $2\alpha\lambda > 1$ so $c_1 < \infty$ and this gives (33). $\square$

The convergence rate of the expression $(\varphi(n))^{-1/2\lambda}$ in (33) improves towards $\varphi(n)^{-\alpha}$ when λ approaches $1/2\alpha$, so asymptotically we prefer to have ϵ' as small as possible. On the other hand the constant c_1 increases rapidly when $\epsilon' \rightarrow 0$. The inequality (33) means that (21) holds for $\epsilon = \tau(\delta) := c_1/(\delta\varphi(n))^{(\alpha-\epsilon')}$. This implies that (22) also holds by Lemma 2.1. By combining Theorem 3.1 with Lemma 2.3, we get the stronger result of Theorem 3.2. A similar inequality was proved in [16], with a slightly different bound on the right of (37).

THEOREM 3.2. *Under the assumptions of Theorem 3.1, if $M_r(f)$ is the median of r QMC estimators obtained with independent draws of $\mathbf{a}$, then*

$$\mathbb{P}\left[\sup_{f \in \mathcal{F}_{\gamma,\alpha}^{(1)}} |M_r(f) - \mu(f)| > \frac{c_1}{(\delta\varphi(n))^{\alpha-\epsilon'}}\right] \leq \psi(r, \delta). \quad (37)$$

PROOF. The result follows by combining (33) with Lemma 2.3. $\square$

In the RQMC setting, we need an inequality as in (24), where the expectation is with respect to both $\mathbf{a}$ and the RS $\mathbf{V}$. For this, we can think of $\mathbf{V}$ as being generated first, and bound the error conditional on $\mathbf{V}$. We can do this because $\mathbf{a}$ and $\mathbf{V}$ are independent random vectors. For a given realization $\mathbf{v}$ of $\mathbf{V}$, shifting the point set by $\mathbf{v}$ modulo 1 is equivalent to replacing $f \in \mathcal{F}_{\gamma,\alpha}^{(1)}$ by the shifted function $f_{\mathbf{v}}$ defined by $f_{\mathbf{v}}(\mathbf{u}) = f(\mathbf{u} \oplus_1 \mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in [0, 1]^s$. Then it suffices to show that $f_{\mathbf{v}} \in \mathcal{F}_{\gamma,\alpha}^{(1)}$ for all $\mathbf{v} \in [0, 1]^s$.

THEOREM 3.3. *For RQMC with a RS, for any smoothness $\alpha > 1/2$, $0 < \epsilon' < \alpha$, $0 < \delta < 1/2$, and choice of weights γ , there is a constant c_1 identical to that of (37) for which*

$$\mathbb{P}\left[e(\mathcal{F}_{\gamma,\alpha}, P_{n,\theta} \oplus_1 \mathbf{V}) > \frac{c_1}{(\delta\varphi(n))^{\alpha-\epsilon'}}\right] = \mathbb{P}\left[e(\mathcal{F}_{\gamma,\alpha}, P_{n,\theta}) > \frac{c_1}{(\delta\varphi(n))^{\alpha-\epsilon'}}\right] \leq \delta. \quad (38)$$

PROOF. It suffices to prove that for any $f \in \mathcal{F}_{\gamma,\alpha}^{(1)}$, we have $e(f, P_{n,\theta} \oplus_1 \mathbf{V}) = e(f, P_{n,\theta})$, and the result then follows from Theorem 3.1. To show this, note that (28) can be rewritten as (see [8]):

$$\mathcal{V}_{\alpha,\gamma}^2(f) = \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^{-1} |\hat{f}(\mathbf{h})|^2 \prod_{j \in \mathbf{u}(\mathbf{h})} |h_j|^{2\alpha}.$$

Also, for any $f \in \mathcal{F}_{\alpha, \gamma}$ and $\mathbf{v} \in [0, 1]^s$, $\hat{f}_{\mathbf{v}}(\mathbf{h}) = \exp\left(2\pi\sqrt{-1}\mathbf{h}^\top \mathbf{v}\right) \hat{f}(\mathbf{h})$ and therefore $|\hat{f}_{\mathbf{v}}(\mathbf{h})|^2 = |\hat{f}(\mathbf{h})|^2$ [21, Theorem 2.3] or [65, Theorem 2.10]. This implies that $\mathcal{V}_{\alpha, \gamma}^2(f_{\mathbf{v}}) = \mathcal{V}_{\alpha, \gamma}^2(f)$, which means that $\mathcal{V}_{\alpha, \gamma}(f) \leq 1$ if and only if $\mathcal{V}_{\alpha, \gamma}(f_{\mathbf{v}}) \leq 1$, and therefore $e^2(f, P_{n, \theta} \oplus_1 \mathbf{v}) = e^2(f_{\mathbf{v}}, P_{n, \theta}) = e^2(f, P_{n, \theta})$. $\square$

As a direct consequence, by combining Theorem 3.3 with Lemma 2.3, we have:

THEOREM 3.4. *Under the assumptions of Theorem 3.1, if $M_r(f)$ is the median of r independent replicates of an RQMC estimator that uses a random $\mathbf{a}$ and a RS, we have*

$$\mathbb{P}\left[\sup_{f \in \mathcal{F}_{\gamma, \alpha}^{(1)}} |M_r(f) - \mu(f)| > \frac{c_1}{(\delta\varphi(n))^{\alpha-\epsilon'}}\right] \leq \psi(r, \delta). \quad (39)$$

By combining Corollary 2.5 with either Theorem 3.2 or Theorem 3.4 with $\beta = \alpha - \epsilon'$, $K = K(n) = c_1(\varphi(n))^{-\beta}$, $\delta_0 < 1/2$, and q and ρ_0 defined in the corollary, we obtain that for either QMC or RQMC,

$$\mathbb{E}[|M_r - \mu|^p] \leq c_1^p (\varphi(n))^{-p\beta} \left[\delta_0^{-p\beta} + 4^{p\beta-1} p\beta \rho_0^{q+1} / (q(1-2\delta_0)) \right] = O((\varphi(n))^{-p\beta} \left[\delta_0^{-p\beta} + O(\rho_0^{q+1}/q) \right]). \quad (40)$$

Again, this bound converges to zero only when $n \rightarrow \infty$ and not when $r \rightarrow \infty$ for n fixed, because $\delta_0^{-p\beta} > 2^{p\beta}$. For the MSE ($p = 2$), it converges as $O(n)^{-2\beta}$, which is almost $O(n)^{-2\alpha}$. In contrast, if we take the average A_r in place of the median M_r , with random generating vectors, the MSE converges to zero with r even for fixed n , but its convergence rate cannot beat $O(rn)^{-1}$, as shown by the following example.

Example 3.5. Consider the one-periodic smooth function $f(\mathbf{u}) = e^{2\pi i \mathbf{h}^\top \mathbf{u}} = e^{2\pi i(u_1 - u_2)}$ for $\mathbf{u} \in [0, 1]^2$, which is the Fourier basis function that corresponds to $\mathbf{h} = (1, -1)$. Note that this function is constant over the main diagonal and also constant over any shift modulo 1 of the main diagonal. Suppose n is prime and we draw $\mathbf{a} = (a_1, a_2)$ at random from $\mathcal{A}_n^2$. With probability $1/n$, we have $a_1 = a_2$, in which case all the lattice points are integer multiples of the vector $(1/n, 1/n)$, so they all belong to the main diagonal. Then, the function f takes the same value at all points of P_n for QMC with this lattice and also the same value at all points of $\tilde{P}_n$ for RQMC with the randomly-shifted lattice. In fact, this function f has a single nonzero Fourier coefficient, equal to 1, for $\mathbf{h} = (1, -1)$. When $a_1 = a_2$, this $\mathbf{h}$ is in the dual lattice, so the absolute QMC error and the RQMC variance are both equal to 1, regardless of the value of n . This means that with probability $1/n$, the MSE of the QMC or RQMC estimator conditional on $\mathbf{a}$ are both equal to 1. Thus, the unconditional $\text{MSE}[X]$ cannot converge faster than $O(1/n)$, so $\text{MSE}[A_r]$ cannot converge faster than $O(1/(rn))$.

3.3 Randomizing the number of points

In a similar setting as in Section 3.1, [25] propose the following QMC and RQMC lattice-rule estimators. Instead of fixing the number of points, they select a *maximum* number of points n , and the actual number of points p is chosen randomly from the set $\mathcal{P}_n$ of prime numbers in $\{[n/2] + 1, \dots, n\}$, for each QMC or RQMC replication. Once p is chosen, the generating vector $\mathbf{a}$ is drawn uniformly from a set $Z_p \subset \{1, \dots, p-1\}^s$ of vectors whose weighted Zaremba index

$$\rho_{\alpha, \gamma}(P_n) = \min_{\mathbf{0} \neq \mathbf{h} \in L_s} \gamma_{\mathbf{u}(\mathbf{h})}^{-1} \prod_{j \in \mathbf{u}(\mathbf{h})} |h_j|^\alpha \quad (41)$$

exceeds a threshold selected so that at least half of the vectors $\mathbf{a}$ are in Z_p . The random parameter here is $\theta = (p, \mathbf{a})$. For a given θ , the QMC estimator is $X = X(f) = \hat{\mu}_{p, \text{qmc}}$ for this (deterministic) lattice rule (no shift). The authors of

[25] proved that with this randomized method, for $\alpha > 1/2$ and weights having the product form $\gamma_{\mathbf{u}} = \prod_{j \in \mathbf{u}} \gamma_j$,

$$\sup_{f \in \mathcal{F}_{\gamma, \alpha}^{(1)}} \mathbb{E}[|X(f) - \mu(f)|] = O(n^{-\alpha-1/2+\epsilon'}) \quad (42)$$

for any $\epsilon' > 0$. They also considered the RQMC estimator $X = \widehat{\mu}_{p, \text{rqmc}}$ obtained by adding a RS and proved that it satisfies (42) (with $\mathbb{E}$ replaced by $\tilde{\mathbb{E}}$) and also the stronger property

$$\sup_{f \in \mathcal{F}_{\gamma, \alpha}^{(1)}} \tilde{\mathbb{E}}[|X(f) - \mu(f)|^2] = O(n^{-2\alpha-1+\epsilon'}) \quad (43)$$

for any $\epsilon' > 0$. These rates are better than those achieved by lattice rules with a fixed number of points. But implementing this approach requires knowing appropriate weights and an appropriate threshold on the Zaremba index.

To remove these requirements, Goda and Krieg [15] considered generating $\mathbf{a}$ uniformly from the whole set $\{1, \dots, p-1\}^s$ instead of from Z_p , and taking the median $M_r(f)$ of r independent replicates of the QMC estimator. Their development is for general weights $\gamma_{\mathbf{u}} \leq 1$, not only product weights, and the algorithm requires no knowledge of the weights. They show that (42) also holds for general weights, and that for $r \geq r_0$ which depends on all relevant parameters,

$$\sup_{f \in \mathcal{F}_{\gamma, \alpha}^{(1)}} \mathbb{E}[|M_r(f) - \mu(f)|] = O(n^{-\alpha-1/2+\epsilon'})$$

for any $\epsilon' > 0$, where the hidden constant depends on ϵ' . They also prove in their Theorem 12 a result similar to the following Theorem 3.6. We provide a much shorter proof based on our previous results. Our Theorem 3.7 extends this result to the RQMC case, i.e., with an additional RS.

THEOREM 3.6. *For any $\alpha > 1/2$, $0 < \epsilon' < \alpha$, and $0 < \delta < 1/2$, there is a constant $c_2 = c_2(s, \alpha, \epsilon')$ independent of n such that*

$$\mathbb{P} \left[\sup_{f \in \mathcal{F}_{\gamma, \alpha}^{(1)}} |M_r(f) - \mu(f)| > \frac{c_2}{(\delta n)^{\alpha-\epsilon'}} \right] \leq \psi(r, \delta). \quad (44)$$

PROOF. For any fixed choice of p , Theorem 3.1 applies with $\varphi(n)$ replaced by $p-1$. Since $p-1 \geq n/2$, this gives

$$\mathbb{P} \left[\sup_{f \in \mathcal{F}_{\gamma, \alpha}^{(1)}} |X(f) - \mu(f)| > \frac{c_1}{(\delta n/2)^{\alpha-\epsilon'}} \right] \leq \sup_{p \in \mathcal{P}_n} \mathbb{P} \left[\sup_{f \in \mathcal{F}_{\gamma, \alpha}^{(1)}} |X(f) - \mu(f)| > \frac{c_1}{(\delta(p-1))^{\alpha-\epsilon'}} \middle| p \right] \leq \delta,$$

where the first probability is with respect to $(p, \mathbf{a})$ and the second probability is with respect to $\mathbf{a}$ conditional on p . Replacing $2^{\alpha-\epsilon'} c_1$ by c_2 in the last equation and applying Lemma 2.3 completes the proof. $\square$

THEOREM 3.7. *If $M_r(f)$ is the median of r RQMC estimators obtained with independent draws of p , $\mathbf{a}$ and $\mathbf{V}$, then the result of Theorem 3.6 also holds with $\mathbb{P}$ replaced by $\tilde{\mathbb{P}}$.*

PROOF. As in the proof of Theorem 3.3, for fixed p , $\mathbf{a}$, and $\mathbf{v}$, the worst-case error $e(f, P_{\theta} \oplus \mathbf{v})$ of the RQMC estimator is the same as the worst-case QMC error $e(f, P_{\theta})$. This implies that (44) holds with $\mathbb{P}$ replaced with $\tilde{\mathbb{P}}$. $\square$

With Theorems 3.6 and 3.7, we get the same bound as in (40), but with $c_1(\varphi(n))^{-\beta}$ replaced by $c_2 n^{-\beta} = 2^{\alpha-\epsilon'} c_1 n^{-\beta}$.

4 DIGITAL NETS

4.1 Basic constructions and properties

A

elements in $\mathbb{Z}_b = \{0, \dots, b-1\}$, $w \geq k$ rows, k columns, and whose first k rows are linearly independent. The QMC point set is $P_n = \{\mathbf{u}_i = (u_{i,1}, \dots, u_{i,s}), i = 0, \dots, n-1\}$, where coordinate j of point i is $u_{i,j} = \sum_{\ell=1}^w u_{i,j,\ell} b^{-\ell}$, whose digits are defined by [11, 37, 51]:

$$\begin{pmatrix} u_{i,j,1} \\ \vdots \\ u_{i,j,w} \end{pmatrix} = C_j \begin{pmatrix} a_{i,0} \\ \vdots \\ a_{i,k-1} \end{pmatrix} \bmod b, \quad \text{with } i = \sum_{\ell=0}^{k-1} a_{i,\ell} b^\ell.$$

The base b is usually a prime number, and the most popular choice is $b = 2$. The linear independence of the first k rows ensures that each projection of P_n over a single coordinate contains exactly one point in each interval $[i/n, (i+1)/n)$, for $i = 0, \dots, n-1$. The choice of the generating matrices determines the uniformity of higher-dimensional projections. A common way to measure this uniformity is by assessing the equidistribution of P_n over partitions of $[0, 1]^s$ into b -adic intervals of the form $I_{\mathbf{q}}(\boldsymbol{\ell}) = \prod_{j=1}^s [\ell_j b^{-q_j}, (\ell_j + 1) b^{-q_j})$, for $\boldsymbol{\ell} = (\ell_1, \dots, \ell_s)$, $0 \leq \ell_j < b^{q_j}$, for a given $\mathbf{q} = (q_1, \dots, q_s) \in \mathbb{N}_0^s$. If for every $\mathbf{q}$ such that $q_1 + \dots + q_s \leq q - t$, there are exactly b^t points from P_n in $I_{\mathbf{q}}(\boldsymbol{\ell})$ for each $\boldsymbol{\ell}$, we say that P_n is a (t, k, s) -net in base b . The smallest t for which is hold is called the t -value of the net, and is often used as a quality parameter [51]. Every projection $P_n(\mathbf{u})$ of P_n onto a subset $\mathbf{u}$ of coordinates also has a t -value, denoted $t_{\mathbf{u}}$, and one can take a weighted sum of these t -values, with weights $\gamma_{\mathbf{u}} \geq 0$, as a more general figure of merit [46].

The best known constructions such as those of Sobol' [66], Faure [13], and Niederreiter [50] are based on non-singular and upper-triangular generating matrices with an infinite number of rows and columns. These infinite generating matrices define an infinite sequence of points. For a digital net with b^k points, the C_j 's are constructed by taking the first k rows and k columns of these matrices. Increasing the number of points from b^k to b^{k+1} amounts to adding one row and one column to each matrix, and increasing the dimension is done by adding new matrices. The infinite matrices are built in a way that for any choice of $k > 0$, the net with 2^k points has a good t -value. Since the t -value depends only on the first k rows of the C_j 's, there is no need to have more than k rows when we use this criterion, and this is why square upper-triangular generating matrices are deemed good enough. On the other hand, for other criteria than the t -value, generating matrices with more than k rows may be preferable [32].

The (popular) Sobol' constructions use $b = 2$. Their infinite generating matrices are defined by *direction numbers* that determine the first few columns, and primitive polynomials over $\mathbb{F}_2$ that determine the further columns via recurrences. Their choice has an impact on the t -values of the Sobol' nets' projections. A popular source for the direction numbers is [24], who selected them by looking at the t -values of the two-dimensional projections. **There is also an important class of digital nets that are defined as lattice rules in a space where the integers are replaced by polynomials with coefficients in the finite field $\mathbb{F}_b$ and the rational numbers are replaced by quotients of such polynomials [11, 35, 44]. Those have generating matrices with more than k rows.**

For RQMC with a digital net, the easiest way to randomize the points without damaging their equidistribution properties is by adding a *random digital shift* (RDS). This is the counterpart of a RS for lattice rules, and it works as follows [11, 28, 33]: generate a single random point $\mathbf{V} = (V_1, \dots, V_s)$ uniformly in $[0, 1]^s$ and add the digits of the base- b expansion of each coordinate V_j to the corresponding digits of each point $\mathbf{u}_i$, modulo b . For $b = 2$, this means xoring

the binary digits of $\mathbf{V}$ with the corresponding binary digits of all the points. We denote the resulting randomized point set by $\tilde{P}_n = P_n \oplus_b \mathbf{V}$. For high-quality generating matrices, with many rows, this should be sufficient.

However, when the initial generating matrices are only $k \times k$, as for the standard point sets, the points after an RDS will differ only in the first k digits of their coordinates. All other digits will be the same as the digits of $\mathbf{V}$. This means that for any projection over a single coordinate j , all the points will be at the same distance from the left boundary in the b -adic intervals $[\ell b^{-k}, (\ell + 1)b^{-k})$ to which they belong. They will be at positions $\Delta, \Delta + 1/n, \dots, \Delta + (n - 1)/n$, where $0 \leq \Delta = (V_j \bmod (1/n)) < 1/n$. Often, the initial generating matrices also have too much structure. For example, the projections of the Sobol' points on certain subsets of coordinates have very poor uniformity [24, 62, 71]. For these reasons, it is common practice to apply deeper randomizations than just an RDS.

The two most popular methods for this are the *left matrix scramble* (LMS) of [47] and the *nested uniform scramble* (NUS) of [54]. These randomizations preserve the t -values of the point set and of its projections. LMS left-multiplies (modulo b) each initial $k \times k$ matrix C_j by a random $w \times k$ matrix L_j whose first k rows form a non-singular lower-triangular submatrix, the entries below the diagonal are independent random digits, and $w \geq k$. This gives another digital net, with new randomized generating matrices. For RQMC, an RDS is still required after the LMS, because each point does not have the uniform distribution after LMS alone (e.g., the point $\mathbf{0}$ is unchanged); see [56, Remark 3.2]. Other linear scrambles of this type are discussed in [48, 56]. In applications, applying LMS before the RDS often reduces the variance by a considerable amount when w is large enough, but has little impact for $w = k$ [32, Section 2.3]. When $w \gg k$, it often has more impact than trying to find better direction numbers for the initial $k \times k$ matrices.

NUS applies random nested permutations to each digit of each coordinate, for all the points [11, 28, 54]. It can be applied to any point set, not only digital nets, and the randomized point set is not a digital net. Applying LMS or NUS is justified by the fact that with these methods, when the integrand f is sufficiently smooth, the variance of the RQMC estimator is known to converge as $O(n^{-3}(\log n)^{s-1})$ [23, 55, 72]. This is under the assumption that the randomized points have an infinite number of output digits, but we can usually observe this rate in practice with fewer bits than machine precision. With an RDS alone, the rate is typically slower, but not always.

LMS and NUS can give this faster rate because LMS returns very good generating matrices with high probability, and NUS gives excellent uniformity with high probability. But there is also a small chance of getting a bad outcome. In particular, for LMS applied to coordinate j , with probability $1/n$, row $k + 1$ of matrix L_j will be all zero, so the $(k + 1)$ -th digit of all the points will be 0 for this coordinate. If we add an RDS, for $b = 2$, all the points will then end up in the same half of the interval $[\ell b^{-k}, (\ell + 1)b^{-k})$ to which they belong for this coordinate. In that case, even for a one-dimensional linear function, the conditional variance with RDS cannot converge faster than $\Omega(n^{-2})$. Since this occurs with probability $1/n$, the unconditional variance with LMS + RDS cannot converge faster than $\Omega(n^{-3})$. Pan and Owen [59] pointed this out and showed that by taking the median of r replicates of the RQMC estimator with LMS + RDS instead of their average, a faster convergence rate can be obtained. They showed that for analytic functions f in one dimension, if $r = \Omega(k)$, the MSE of the median converges at a super-polynomial rate, and if $r = \Omega(k^2)$, then $\text{MSE}(\text{median}) \approx O(n^{-0.42c \log_2(n)})$. This was the first paper to suggest taking the median instead of the average in RQMC. In [60], they extended their results to $s \geq 1$ dimensions, showing that $\text{MSE}(\text{median}) \approx O(n^{-0.42c \log_2(n)/s})$. Further results for non-analytic functions f are obtained in [19, 58] and are discussed later in this section.

Note that using the initial $k \times k$ generating matrices directly without LMS or NUS is equivalent to having the first k rows of each L_j forming the identity and all the other rows equal to zero. By the argument of the previous paragraph, this means that for a non-constant linear function, the variance cannot converge faster than $\Omega(n^{-2})$. On

the other hand, it can converge faster for smooth one-periodic functions. To provide insight on why this happens, let us consider a one-dimensional function $f : [0, 1] \rightarrow \mathbb{R}$ for which $\sup_{0 \leq u \leq 1} |f''(u)| \leq K$ and suppose we estimate the integral by using the point set $P_n(\Delta) = \{\Delta, \Delta + 1/n, \dots, \Delta + (n-1)/n\}$ for $0 \leq \Delta < 1/n$. For $\Delta = 1/(2n)$, this gives the well-known *midpoint rule*, for which the absolute integration error is bounded by $K/(24n^2)$. This method is equivalent to approximating f by a piecewise-linear function f_{pl} with n linear pieces over equal intervals, and integrating f_{pl} exactly. For $\Delta \neq 1/(2n)$, if s_i is the slope of the linear piece over $[i/n, (i+1)/n]$, the additional error on that piece is $(\Delta - 1/(2n))s_i/n$, and then the integration error on f_{pl} is $(\Delta - 1/(2n)) \sum_{i=0}^{n-1} s_i/n = (\Delta - 1/(2n))(f(1) - f(0))$. In general, this error is $O(n^{-1})$, but if $f(1) = f(0)$, it vanishes for any Δ , and the integration error on f becomes the same as with the midpoint rule, which is $O(n^{-2})$. This means that for either a lattice rule with a RS or for Sobol' points with a RDS, if f is a smooth one-periodic function, the RQMC variance is $O(n^{-4})$. When the function is sufficiently smooth but not one-periodic, it suffices to apply the baker transformation to get the $O(n^{-4})$ rate. This simple analysis extends to higher dimensions, although it then requires more sophisticated tools and it is shown that for any $\epsilon > 0$, there are digital nets for which the worst-case error with a RDS is $O(n^{-2+\epsilon})$, so the variance is $O(n^{-4+\epsilon})$, for smooth one-periodic functions [3]. This aspect is explored further in [36]. In that case, applying LMS or NUS may increase the variance and even degrade the rate compared to using a RDS alone. This occurs in our numerical examples.

When we opt for scrambling the points, to avoid the bad random scrambles, we could think of constructing directly better generating matrices with $w > k$ rows, by minimizing a *figure of merit* (FOM) similar to the $\mathcal{P}_{2\alpha, \gamma}(P_n)$ defined in (29). Such figures do exist for digital nets and are defined by using a similar development as for lattice rules, with the Fourier expansion replaced by a Walsh expansion [11, 33, 44]. If f has a Walsh expansion with coefficients $\hat{f}(\mathbf{k})$ for $\mathbf{k} = (k_1, \dots, k_s) \in \mathbb{N}_0^s$, the QMC error with a digital net in base b is $\sum_{\mathbf{0} \neq \mathbf{k} \in \mathcal{D}} \hat{f}(\mathbf{k})$ if this sum converges absolutely, and the RQMC variance with a RDS is $\sum_{\mathbf{0} \neq \mathbf{k} \in \mathcal{D}} |\hat{f}(\mathbf{k})|^2$, where $\mathcal{D} \subset \mathbb{N}_0^s$ is the dual net. It was shown in [5–7] and [9, middle of page 2687] that if f is in a weighted space of smoothness $\alpha \geq 2$, which means (roughly) that its mixed partial derivatives of order up to α with respect to each coordinate are integrable, then

$$|\hat{f}(\mathbf{k})| = O\left(\mathcal{V}_{\gamma, \alpha, q}(f) \gamma_{\mathbf{u}}(\mathbf{k}) b^{-\mu_{\alpha}(\mathbf{k})}\right), \quad (45)$$

where the variation $\mathcal{V}_{\gamma, \alpha, q}(f)$ is defined in (46) and $\mu_{\alpha}(\mathbf{k})$ depends on the digits of $\mathbf{k}$ in base b . In principle, one can then select a good LMS based on an FOM of the form $\sum_{\mathbf{0} \neq \mathbf{k} \in \mathcal{D}} \gamma_{\mathbf{u}}(\mathbf{k}) b^{-\mu_{\alpha}(\mathbf{k})}$ for QMC and $\sum_{\mathbf{0} \neq \mathbf{k} \in \mathcal{D}} \gamma_{\mathbf{u}}^2(\mathbf{k}) b^{-2\mu_{\alpha}(\mathbf{k})}$ for RQMC, or an approximation of those, as further described in [17, 32, 49]. This requires knowledge of appropriate weights and significant work.

4.2 The median estimator for digital nets with a random LMS

As we saw, a common way of using digital nets for RQMC is to start with $k \times k$ generating matrices with good properties, then apply a random LMS with more than k rows, and add a RDS. The justification is that the standard FOMs for digital nets depend only on the first k rows, whereas better convergence results are often achieved with more rows [32]. In this setting, the random LMS matrices L_j play the same role as the random generating vectors $\mathbf{a}$ for rank-1 lattice rules. Usually, r independent replicates of this LMS + RDS estimator are obtained and μ is estimated by their average. To reduce the impact of the rare bad LMS choices, one can replace the average by the median as in (12).

As mentioned earlier the first results exploiting this idea were obtained for analytical functions. Here we present results obtained on the spaces of functions mentioned in the previous section, where the parameter α quantifies the level of smoothness. Namely, Goda et al. [19] study the median estimator in weighted Sobolev spaces of order α as

introduced in [9]. For given integer $\alpha \geq 2$, set of weights γ , and norm parameter $q \in [1, \infty)$, consider the variation

$$V_{\gamma, \alpha, q}(f) = \sup_{\mathbf{u}' \subset \{1, \dots, s\}} \gamma_{\mathbf{u}'}^{-1} \left(\sum_{\mathbf{u} \subset \mathbf{u}'} \sum_{\tau_{\mathbf{u}' \setminus \mathbf{u}} \in \{1, \dots, \alpha\}^{|\mathbf{u}' \setminus \mathbf{u}|}} \int_{[0,1]^{|\mathbf{u}'|}} \left| \int_{[0,1]^{s-|\mathbf{u}'|}} f^{(\tau_{\mathbf{u}' \setminus \mathbf{u}}, \alpha_{\mathbf{u}}, 0)}(\mathbf{u}) d\mathbf{u}_{-\mathbf{u}} \right|^q d\mathbf{u}_{\mathbf{u}} \right)^{1/q}, \quad (46)$$

where $(\tau_{\mathbf{u}' \setminus \mathbf{u}}, \alpha_{\mathbf{u}}, 0)$ denotes the vector $\mathbf{k} \in \mathbb{N}_0^s$ such that $k_j = \tau_j \in \{1, \dots, \alpha\}$ if $j \in \mathbf{u}' \setminus \mathbf{u}$, $k_j = \alpha$ if $j \in \mathbf{u}$, and $k_j = 0$ otherwise. Wherever $\gamma_{\mathbf{u}'} = 0$ we set the contribution from the inverse weight and corresponding integral to 0 by convention ($0/0 = 0$). We denote the corresponding Banach space of functions by $\mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}$.

By using the bound (45) on the Walsh coefficients, the worst-case error for this space can be bounded by the following quantity that involves the $\mu_{\alpha}(\mathbf{k}_{\mathbf{u}})$ and is independent of q :

$$e(\mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}, P_n) \leq \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}} C_{\alpha}^{|\mathbf{u}|} \sum_{\mathbf{k}_{\mathbf{u}} \in \mathcal{D}_{\mathbf{u}}} b^{-\mu_{\alpha}(\mathbf{k}_{\mathbf{u}})},$$

where the constants C_{α} only depend on b and α and $\mathcal{D}_{\mathbf{u}}$ is the dual of the net $P_n(\mathbf{u})$. While this bound holds for any digital net assumed to have $ak \times k$ generating matrices, the bound given in the next theorem requires that the underlying digital net P_n to which random scrambles are applied has projections $P_n(\mathbf{u})$ with t -values $t_{\mathbf{u}}$ that satisfy

$$t_{\mathbf{u}} \leq \sum_{j \in \mathbf{u}} (\log_b j + \log_b \log_b(j+b) + 1) \quad \text{for every } \mathbf{u} \subset \{1, \dots, s\}. \quad (47)$$

Niederreiter sequences do satisfy this bound, while for Sobol' sequences, we have the slightly larger bound [70]:

$$t_{\mathbf{u}} \leq \sum_{j \in \mathbf{u}} (\log_2 j + \log_2 \log_2(j+1) + \log_2 \log_2 \log_2(j+3) + C) \quad \text{for every } \mathbf{u} \subset \{1, \dots, s\}, \quad (48)$$

where C is a constant. The following is Theorem 3.10 in [19]. We give a shorter proof using our framework of Section 2.

THEOREM 4.1. *Let s, α, γ, q be any choice of valid parameters to define $\mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}$. For a given $n = b^k$, let $P_{n, \theta}$ be a digital net satisfying (47), scrambled by randomly chosen $\infty \times k$ matrices $\theta = (L_1, \dots, L_s)$. Then for any $\delta \in (0, 1)$,*

$$\mathbb{P} \left[e(\mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}, P_{n, \theta}) > \varepsilon_n \right] \leq \delta,$$

where

$$\varepsilon_n = \inf_{\lambda \in (1/\alpha, 1]} \left(\frac{1}{\delta b^k} \sum_{\emptyset \neq \mathbf{u} \subset \{1, \dots, s\}} \gamma_{\mathbf{u}}^{\lambda} \left(\frac{b^2}{b-1} \right)^{|\mathbf{u}|} C_{\alpha}^{\lambda |\mathbf{u}|} A_{\alpha, \lambda}^{|\mathbf{u}|} \prod_{j \in \mathbf{u}} j \log_b(j+b) \right)^{1/\lambda} \quad (49)$$

and $A_{\alpha, \lambda} = \sum_{k=1}^{\infty} b^{-\lambda \mu_{\alpha}(k)}$.

PROOF. For any $\lambda \in (1/\alpha, 1]$:

$$\begin{aligned} \mathbb{E}[e^{\lambda}(\mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}, P_{n, \theta})] &\leq \frac{1}{b^k} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^{\lambda} C_{\alpha}^{\lambda |\mathbf{u}|} \left(\frac{b}{b-1} \right)^{|\mathbf{u}|} b^{t_{\mathbf{u}}} \sum_{\mathbf{k}_{\mathbf{u}} \in \mathbb{N}^{|\mathbf{u}|}} b^{-\lambda \mu_{\alpha}(\mathbf{k}_{\mathbf{u}})} \\ &\leq \frac{1}{b^k} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^{\lambda} C_{\alpha}^{\lambda |\mathbf{u}|} \left(\frac{b}{b-1} \right)^{|\mathbf{u}|} b^{t_{\mathbf{u}}} A_{\alpha, \lambda}^{|\mathbf{u}|} \\ &\leq \frac{1}{b^k} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^{\lambda} C_{\alpha}^{\lambda |\mathbf{u}|} \left(\frac{b}{b-1} \right)^{|\mathbf{u}|} A_{\alpha, \lambda}^{|\mathbf{u}|} \prod_{j \in \mathbf{u}} j \log_b(j+b), \end{aligned}$$

where the last inequality follows from (47). Markov's inequality yields the result. $\square$

Under some conditions on the weights γ that depend on the bounds on the t_u 's, one has $\varepsilon_n \in O(n^{-(\alpha-\epsilon')})$. Prior to this result, Goda and L'Ecuyer [16] studied that same space of functions with higher-order polynomial lattice rules and obtained a concentration inequality with similar behavior, i.e., with $\varepsilon_n \in O(n^{-(\alpha-\epsilon')})$. If we add a RDS V after the random LMS to get an unbiased RQMC estimator, we have the following.

THEOREM 4.2. *Let $V \sim [0, 1]^s$ be a random vector. Then with all of the same hypotheses as Theorem 4.1,*

$$\tilde{\mathbb{P}} \left[e(\mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}, P_{n, \theta} \oplus_b V) > \varepsilon_n \right] \leq \delta,$$

for the same values of δ and ε_n .

PROOF. From [9, Remark 3.1], the bound on the worst-case error derived for the (unshifted) digital net also holds for a digitally shifted net. Therefore

$$\sup_{\mathbf{v} \in [0, 1]^s} \mathbb{E} \left[e(\mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}, P_{n, \theta} \oplus_b \mathbf{v}) \right] \leq \frac{1}{b^k} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^\lambda C_\alpha^{\lambda|\mathbf{u}|} \left(\frac{b}{b-1} \right)^{|\mathbf{u}|} A_{\alpha, \lambda}^{|\mathbf{u}|} \prod_{j \in \mathbf{u}} j \log_b(j+b),$$

and thus the same steps as in the deterministic case yield the desired result. $\square$

By combining Theorem 4.4 with Lemma 2.3, we obtain the following result for the median.

THEOREM 4.3. *Under the assumptions of Theorem 4.2, if $M_r(f)$ is the median of r RQMC estimators obtained with independent draws of $L_1, \dots, L_s$ and independent RDSs, then for any $0 < \delta < 1/2$ and with ε_n given by (49), we have*

$$\mathbb{P} \left[\sup_{f \in \mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}, V_{\alpha, \gamma}(f) \leq 1} |M_r(f) - \mu(f)| > \varepsilon_n \right] \leq \psi(r, \delta). \quad (50)$$

By taking the limit $\alpha \rightarrow \infty$, one obtains a space of infinitely smooth functions, for which results on the median are presented in [19]. A less restrictive space of functions is the weighted Sobolev space of first order, denoted $\mathcal{F}_{1, \gamma}^{\text{sob}}$, which corresponds to taking $\alpha = q = 1$ in $\mathcal{F}_{\alpha, \gamma, q}^{\text{sob}}$. It is based on the variation

$$V_{\gamma, 1}(f) = \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^{-1} \int \left| \frac{\partial^{|\mathbf{u}|}}{\partial \mathbf{x}_{\mathbf{u}}} f(\mathbf{x}_{\mathbf{u}}, \mathbf{1}) \right| d\mathbf{x}_{\mathbf{u}}.$$

Goda et al. [19] obtain a concentration inequality for a digital net, which is again scrambled by randomly chosen matrices $\theta = (L_1, \dots, L_s)$. The Walsh coefficients for functions in that space do not obey bounds such as (45), valid for $\alpha \geq 2$. For this reason, the worst-case error must be analyzed using the more general bound

$$e(\mathcal{F}_{1, \gamma}^{\text{sob}}, P_n) \leq D_{\gamma}^*(P_n) := \sup_{\mathbf{y} \in [0, 1]^s} \max_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}} |\Delta(P_n, (\mathbf{y}_{\mathbf{u}}, \mathbf{1}))|,$$

where $\Delta(P_n, \mathbf{y}) = (1/n) \sum_{i=1}^n \mathbb{I}_{\mathbf{u}_i \in [0, \mathbf{y}]} - |y_1 \dots y_s|$ is the local discrepancy over $[0, \mathbf{y}]$. Theorem 3.4 of [19] says that for any $\delta \in (0, 1)$ and any digital net satisfying (47), we have $\mathbb{P} \left[e(\mathcal{F}_{1, \gamma}^{\text{sob}}, P_{n, \theta}) > \varepsilon_n \right] \leq \delta$, where

$$\varepsilon_n = \frac{3}{\delta n} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}} \prod_{j \in \mathbf{u}} \left[1 + m \frac{b(b+1)}{3} j \log_b(j+b) \right]. \quad (51)$$

The next theorem extends this result to a digital net with a RDS.

THEOREM 4.4. *Let $V \sim [0, 1]^s$ be a random vector. For a given $n = b^k$, let $P_{n,\theta}$ be a digital net satisfying (47) linearly scrambled by randomly chosen matrices $\theta = (L_1, \dots, L_s)$ of size $\infty \times k$. Then for any $\delta \in (0, 1)$, with ε_n given by (51),*

$$\mathbb{P} \left[e(\mathcal{F}_{1,\gamma}^{\text{sob}}, P_{n,\theta} \oplus V) > \varepsilon_n \right] \leq \delta.$$

PROOF. We proceed by showing that the bound on $\mathbb{E}(D_Y^*(P_{n,\theta}))$ established in the proof of [19, Thm 3.14] also holds for $\tilde{\mathbb{E}}(D_Y^*(P_{n,\theta} \oplus V))$. This establishes a bound of the form (24), so the approach outlined in Section 2.2 can be used. The key is to consider the m -truncated version of a point set P_n , denoted $P_n^{(m)}$, obtained by only keeping the first m digits of all points, i.e., by setting $u_{i,j,w} = 0$ for all $w > m$ in the base b expansion of its points u_i . As in [19], we also work with the version of the weighted star discrepancy that sums over all $\mathbf{u}$ instead of taking the maximum, i.e., $\tilde{D}_Y^*(P_n) := \sup_{\mathbf{y} \in [0,1]^s} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}} |\Delta(P_n, (\mathbf{y}_{\mathbf{u}}, 1))|$, and then write

$$\tilde{\mathbb{E}}(D_Y^*(P_{n,\theta} \oplus V)) \leq \tilde{\mathbb{E}}(\tilde{D}_Y^*(P_{n,\theta} \oplus V)^{(m)}) + \left[\tilde{\mathbb{E}}(\tilde{D}_Y^*(P_{n,\theta} \oplus V)) - \tilde{\mathbb{E}}(\tilde{D}_Y^*(P_{n,\theta} \oplus V)^{(m)}) \right]. \quad (52)$$

To deal with the first term, we use [11, Thm 3.28 & Cor.A.7], write $(P_{n,\theta} \oplus V)^{(m)} = \{u_i^{(m)}, i = 0, \dots, n-1\}$ and get

$$\begin{aligned} \tilde{D}_Y^*((P_{n,\theta} \oplus V)^{(m)}) &\leq \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}} \left[1 - \left(1 - \frac{1}{b^m}\right)^{|\mathbf{u}|} + \sum_{\mathbf{k} \in \mathbb{N}_0^{|\mathbf{u}|}, \|\mathbf{k}\|_{\infty} < b^m} \left(\prod_{j \in \mathbf{u}} \tilde{r}_b(k_j) \right) \left| \frac{1}{n} \sum_{i=0}^{n-1} \text{wal}_{\mathbf{k}}(u_i^{(m)}) \right| \right] \\ &= \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}} \left[1 - \left(1 - \frac{1}{b^m}\right)^{|\mathbf{u}|} + \sum_{\mathbf{k} \in \mathcal{D}_{\mathbf{u}}^{(m)}, \|\mathbf{k}\|_{\infty} < b^m} \left(\prod_{j \in \mathbf{u}} \tilde{r}_b(k_j) \right) \right], \end{aligned}$$

where $\mathcal{D}_{\mathbf{u}}^{(m)}$ is the dual net of the unshifted, m -truncated point set, $\tilde{r}_b(k) = b^{-a_v} (\sin(\pi \kappa_{a_v}/b))^{-2}$ if $k \neq 0$ and $\tilde{r}_b(0) = 1$, and $\text{wal}_{\mathbf{k}}(x)$ is the Walsh basis function in base b with wave number $\mathbf{k}$. This is the same bound as the one obtained in the proof of [19, Thm 3.14], i.e., without the RDS. Therefore their bound on the expected value of this truncated discrepancy also applies in our case and yields $\tilde{\mathbb{E}}(\tilde{D}_Y^*(P_{n,\theta} \oplus V)^{(m)}) \leq 2\delta\varepsilon_n/3$, where ε_n is defined in (51). To deal with the second term in (52), it suffices to use [11, Thm 3.14], which applies here since $u_{i,j} - u_{i,j}^{(m)} \leq b^{-m}$ also holds for points u_i and $u_i^{(m)}$ from $P_{n,\theta} \oplus V$ and $(P_{n,\theta} \oplus V)^{(m)}$, respectively. Hence we can proceed as in [19] and get $\tilde{\mathbb{E}}(\tilde{D}_Y^*(P_{n,\theta} \oplus V)) - \tilde{\mathbb{E}}(\tilde{D}_Y^*(P_{n,\theta} \oplus V)^{(m)}) \leq \delta\varepsilon_n/3$. Putting these two pieces together establishes a bound of the form (24) with $\lambda = 1/2$ and $b(n, 1/2)$ given by ε_n . Using the approach from Section 2.2 based on Markov inequality, we get a bound of the form (25), which is our desired result. $\square$

THEOREM 4.5. *Under the assumptions of Theorem 4.4, if $M_r(f)$ is the median of r QMC estimators obtained with independent draws of $L_1, \dots, L_s$ and independent RSSs, then for any $0 < \delta < 1/2$ we have*

$$\mathbb{P} \left[\sup_{f \in \mathcal{F}_{1,\gamma}, V_{\gamma,1}(f) \leq 1} |M_r(f) - \mu(f)| > \varepsilon_n \right] \leq \psi(r, \delta), \quad (53)$$

with ε_n given by (51).

PROOF. It suffices to combine Theorem 4.4 with Lemma 2.3. $\square$

As a final remark, we note that if the digital net comes from a Sobol' sequence and thus the looser bound (48) holds instead of (47), then all the above results hold with a slightly larger ε_n in which the term $j \log_b(j + b)$ in the product

over $j \in \mathfrak{u}$ must be replaced by $j \log_2(j+1) \log_2 \log_2(j+3)$. Accordingly, the conditions on the weights $\gamma_{\mathfrak{u}}$ required to get $\epsilon_n \in O(n^{-(\alpha-\epsilon')})$ (see, e.g., [19, Corollaries 3.5 and 3.11]) must be slightly modified.

5 NUMERICAL EXPERIMENTS AND ILLUSTRATIONS

5.1 Experimental setting

In this section, we report the results of numerical experiments to examine the distribution of the QMC or RQMC estimator X and compare $\text{MSE}[M_r]$ with $\text{MSE}[A_r]$ in various settings, including when some parameters of the QMC point sets are drawn at random versus when they are fixed. The selected point sets and integrands are similar to those in [41]. We classify the point set constructions in three categories. They are listed below, each one with an identifier used in the text, figures, and the data file names.

(a) Fixed choice of parameters, unbiased estimators (RQMC):

- Lat-RS: lattice rule with RS;
- Lat-RSB: lattice rule with RS followed by baker transform;
- Sob-RDS: Sobol' net with RDS;
- Sob-RDSB: Sobol' net with RDS followed by baker transform;
- Sob-NUS: Sobol' net with NUS;

(b) Random parameters, unbiased estimators (RQMC):

- Lat-RvRS: lattice rule with random α and RS;
- Lat-RvRSB: lattice rule with random α and RS followed by baker transform;
- Lat-RpvRS: lattice rule with random number of points, random α , and RS;
- Lat-RpvRSB: lattice rule with random number of points, random α , RS, and baker;
- Sob-LMS-RDS: Sobol' net with random LMS followed by RDS;

(c) Random parameters, no shift (QMC):

- Lat-Rv: lattice rule with random α ;
- Lat-Rpv: lattice rule with random number of points and random α ;
- Sob-LMS: Sobol' net with random LMS, no shift;

For Lat-RS and Lat-RSB, we used a generating vector α given in [31] and found by LatNet Builder [37, 39] with the $\mathcal{P}_\alpha$ criterion with $\alpha = 2$, product weights with weight $\gamma_j = 2/(2+j)$ for coordinate j , for $n = 2^k$ for all $k = 8, \dots, 18$, with a component-by-component construction in up to 32 dimensions. For the Sobol' nets, we used the initial generating matrices from [24]. For the Sobol' points, the parameters drawn at random in (b) and (c) are the L_j matrices for the LMS. For the QMC setting (c), there is no RS or RDS, and also no baker transformation. We tried all these point sets with $n = 2^k$ points for $k = 10, 12, 14, 16$.

We report our results for the following integrands f , in $s = 2, 4, 8, 16, 32$ dimensions:

$$\begin{aligned}
\text{SmoothPerB4} : \quad f(\mathbf{u}) &= -1 + \prod_{j=1}^s [1 + (1/j)(30u_j^2(1-u_j)^2 - 1)] \\
\text{SumUeU} : \quad f(\mathbf{u}) &= -s + \sum_{j=1}^s u_j \exp(u_j) \\
\text{MC2} : \quad f(\mathbf{u}) &= -1 + (s-1/2)^{-s} \prod_{j=1}^s (s - u_j) \\
\text{SmoothGauss} : \quad f(\mathbf{u}) &= -\Phi(1/\sqrt{2}) + \Phi\left(1 + s^{-1/2} \sum_{j=1}^s \Phi^{-1}(u_j)\right) \\
\text{PieceLinGauss} : \quad f(\mathbf{u}) &= \max\left(s^{-1/2} \sum_{j=1}^s \Phi^{-1}(u_j) - \tau, 0\right) - \varphi(\tau) + \tau\Phi(-\tau) \\
\text{IndSumNormal} : \quad f(\mathbf{u}) &= -\Phi(-\tau) + \mathbb{I}\left[s^{-1/2} \sum_{j=1}^s \Phi^{-1}(u_j) \geq \tau\right]
\end{aligned}$$

where Φ is the cdf of the standard normal distribution and $\mathbb{I}$ is the indicator function. `SmoothPerB4` is the worst-case function in the Korobov space of smoothness $\alpha = 4$ described in Section 3.1, whose square discrepancy is $\mathcal{P}_{8,\gamma}$ with product weights $\gamma_u = \prod_{j \in u} \gamma_j$ and $\gamma_j = 1/j^2$ [20, 38]. Note that $B_4(u) = u^2(1-u)^2 - 1/30$ is the fourth Bernoulli polynomial and `SmoothPerB4` is a weighted sum of Bernoulli polynomials. This function is very smooth and one-periodic, so it does not require a baker transformation and we should observe a fast convergence rate with good lattice rules. The other functions are taken from [41]. None is one-periodic. `SumUeU` is smooth and additive (it is a sum of one-dimensional functions), so its integration error should not depend much on s . `MC2` and `SmoothGauss` are also smooth and bounded. `MC2` has low effective dimension because the coefficients $(s-1/2)^{-s}$ decrease rapidly with s . `SmoothGauss` is increasing and takes its values in the interval $(-a, 1-a)$ where $a = \Phi(1/\sqrt{2})$. `PieceLinGauss` is continuous but not smooth and `IndSumNormal` is discontinuous. Both have infinite variation in the sense of Hardy and Krause [41]. All these test functions have been centered (by adding a constant) so that $\mu = \int_{(0,1)^s} f(\mathbf{u}) d\mathbf{u} = 0$. We pretend μ is unknown and we try to estimate it by RQMC or by QMC with random parameters. For this, we make r independent replicates of the estimator and take the average or the median.

5.2 Exploring and Estimating the Distributions of QMC and RQMC Estimators

In a first stage, we want to discover what types of distributions these QMC/RQMC estimators X can have, for various types of integrands and point sets. This is important because whether M_r is better than A_r as an estimator of the mean depends only on the distribution of X . For each combination of QMC/RQMC method, function f , point set size $n = 2^k$, and dimension s , we generated $N = 10^4$ independent realizations $X_1, \dots, X_N$ of the QMC or RQMC estimator X and saved the N sorted values in a data file. This gives an empirical distribution of the QMC/RQMC estimator which estimates its true distribution for that case. We also made a histogram of these values to visualize the distribution, and we computed the mean, variance, skewness, and excess kurtosis of each sample. We have 6 integrands, 13 types of point sets, 5 values of s , and 4 values of k , for a total of 1560 cases. Results for $s = 1$ and for additional functions are reported in [31]. The total number of integrand evaluations required for this entire experiment was about 612 billions. The computations were using the packages `hups64` and `mcqmcTools` from the SSJ Java library [29].

Recall that if X has mean $\mu' = \mathbb{E}[X]$, then it has variance $\sigma^2 = \mathbb{E}[(X - \mu')^2]$, skewness $\gamma = \mathbb{E}[(X - \mu')/\sigma]^3$, kurtosis $\kappa = \mathbb{E}[(X - \mu')/\sigma]^4 \geq 1 + \gamma^2$, and *excess kurtosis* $\kappa' = \kappa - 3$. Our mean and variance estimators are unbiased, but the reported skewness and kurtosis are just those of the empirical distribution. These estimators can be noisy, especially

when the kurtosis is large. In that case, even if the distribution is symmetric so the true skewness is zero, and even if $N = 10^4$ is quite a large sample size, since the observations far in the tails are rare, we sometimes by chance have one or two observations very far in the tail on one side and none on the other side, or a few more on one side than the other side, and then get a large empirical skewness. This type of behavior occurs occasionally in our data. We recommend looking at the plots and results in [31] while reading the following.

We summarize what we found from this experiment. Many data sets (and histograms) have a high kurtosis, especially for the methods in Category (c). The largest one is around 10000. One can get an idea of the overall skewness and kurtosis distributions from Figure 3. For the lattice point sets, both $\text{Var}[X]$ and the kurtosis are usually larger with the random coefficients in (b) than with the fixed ones in (a). The variance is often larger by a factor between 1 and 100 and sometimes much larger, e.g., for SmoothPerB4 and for MC2 with baker. One exception is SumUeU, where it makes no difference, because it is a sum of one-dimensional functions, so all admissible vectors $\mathbf{a}$ are equivalent. This implies that for Lat-RV, the true variance is exactly zero and the true kurtosis and absolute skewness are undefined, because $\sigma = 0$. Apart from this special case, with random coefficients, taking a fixed or random number of points p makes little difference in the variance and kurtosis of X . Adding the baker transformation improves the convergence rate of the variance from $O(n^{-2})$ to $O(n^{-4})$ for smooth non-periodic functions with the fixed coefficients. It reduces $\text{Var}[X]$ significantly for MC2 and SumUeU, and moderately for SmoothGauss. For SmoothPerB4, the baker transformation brings no improvement because this function is already periodic. It also has much less effect with the random coefficients, because in that case most of the variance is due to the randomness of the coefficients.

For the Sobol' points, NUS and RDS alone usually give a distribution close to normal, with rare exceptions, e.g., with SmoothPerB4 where RDS alone sometimes gives a multimodal distribution. Although Sob-LMS-RDS is known to give the same variance as NUS [71], it often gives a much larger kurtosis, for example for SumUeU, which corresponds exactly to the type of function studied in [59]. The variance of Sob-RDS is much larger than that of Sob-NUS and Sob-LMS-RDS for MC2 and SumUeU, smaller for SmoothPerB4 due to the one-periodicity, and about the same in the other cases, presumably because of the larger effective dimension. Sob-RDSB performs extremely well for non-periodic smooth functions, similar to Lat-RSB, for reasons explained in [36]. It also gives very bizarre histograms in general, often with two or many narrow peaks, sometimes with no density in the center.

Figure 1 shows the histograms we got with the RQMC methods without baker in (a) and (b), for the SmoothPerB4 function, with $s = 8$ and $k = 16$. Adding baker to this one-periodic function gives almost the same histograms, with a slightly larger variance. Taking random lattice parameters gives much larger empirical variance and kurtosis than for the other methods. This behavior often occurs for other functions as well. A large kurtosis is also often observed for Sobol' points with LMS, for other functions. This histogram for Sob-RDS has a funny shape, unfavorable for the median, because the empirical median will always be in one of the two bumps, never in the center. The density seems to be zero around 0, in which case $\text{MSE}[M_r]$ does not converge to 0 when $r \rightarrow \infty$ for n fixed. We have observed this type of histogram in a few cases, mostly with Sob-RDS and Sob-RDSB. **We also found variants with more bumps, often with a bump in the center as in Figure 2 and sometimes not. When the central bump is large, this favors the median.**

To investigate the accuracy of those histograms with $N = 10^4$ as representations of the true distribution of X , we tried in certain cases a much larger sample size of $N = 10^6$ to see how the histograms change, especially when the empirical skewness and kurtosis are large. We found from these experiments that in most RQMC cases, the large empirical skewness in our data is due to statistical noise. With the larger sample size, the two far tails usually become balanced and the empirical skewness gets close to zero, so the true distribution of X appears symmetric, while the kurtosis often remains large. This occurs for example for the Lat-RvRS and Lat-RpvRS cases of Figure 1. Histograms

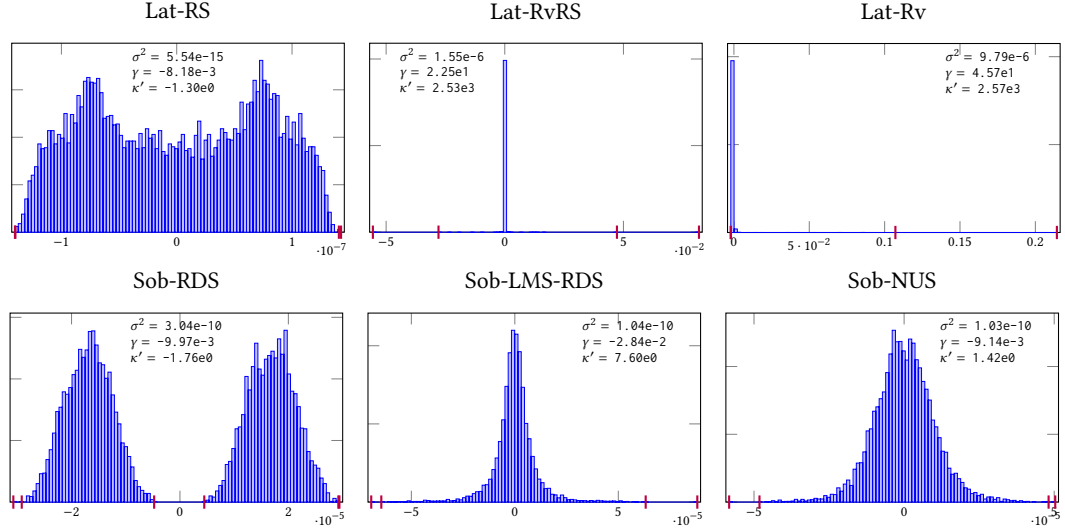


Fig. 1. Histograms of 10,000 realizations of X for six methods for the function SmoothPerB4 with $s = 8$ and $k = 16$. The histograms in the first row are for lattice points and those in the second row are for Sobol' points. All observations are in the range of the horizontal axis, so the pictures with a very narrow peak indicate that most observations are in this narrow area, with a few outliers up to both ends of the axis, which contribute significantly to the high empirical kurtosis. Small red-purple bars on the horizontal axis mark the two smallest and two largest observations. For Sob-RDS, the two red-purple bars between the bumps mark the observations that are closest to zero on each side; there is nothing in between. The two bars for Lat-RS and the last one for Sob-RDS are thicker because each one represents two outliers that are close to each other.

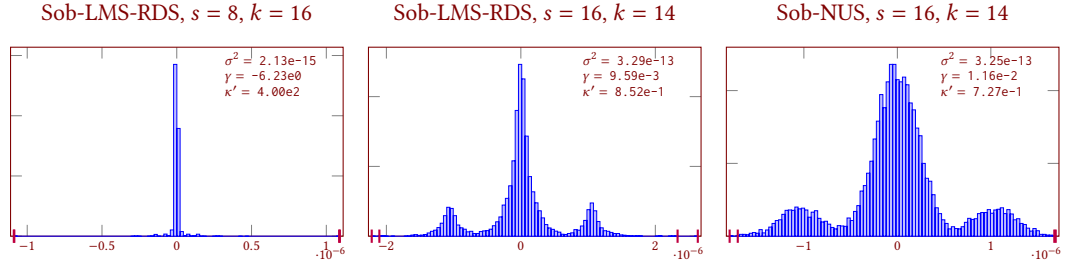


Fig. 2. Histograms of 10,000 realizations of X for the function MC2.

with $N = 10^6$ are given in [31] to illustrate this aspect. In general they do not behave much differently than those with $N = 10^4$. There are also cases where the distribution is slightly asymmetric, usually when k is small. For the QMC methods, most of the distributions are asymmetric, sometimes highly asymmetric, typically with a positive skewness.

5.3 Average vs median to estimate the mean

The second stage of our experiment was to compare A_r vs M_r as estimators of the true mean μ . For each of the 1560 cases, we drew $m = 10000$ independent (bootstrap) subsamples of size $r = 11$ from the data file, with replacement, computed the median M_r for each sample, and took the average of their square to estimate $\text{MSE}[M_r]$. We estimate $\text{MSE}[A_r] = \text{Var}[X]/r + \mathbb{E}^2[X]$ by using the empirical values in the latter expression. Figure 3 shows the kurtosis κ vs

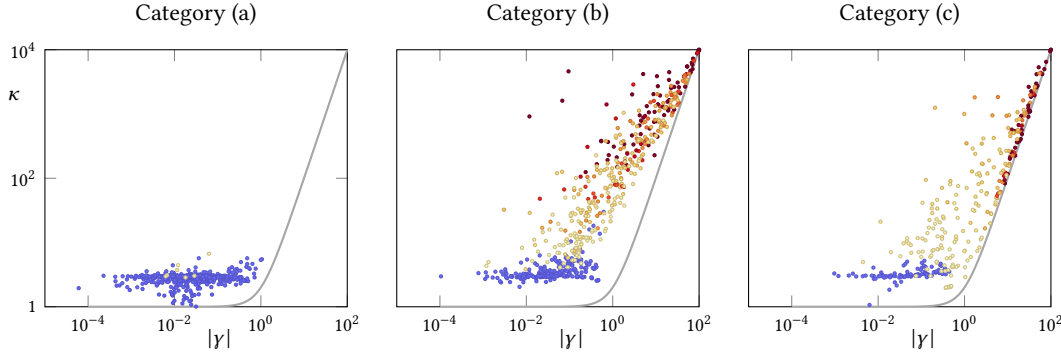


Fig. 3. Plots in log scale of the kurtosis κ vs the absolute skewness $|\gamma|$ of X for the three categories of methods: (a) fixed-parameter RQMC (left); (b) random-parameter RQMC (center); (c) random-parameter QMC (right). The gray line indicates $\kappa = 1 + \gamma^2$; pairs $(|\gamma|, \kappa)$ below this line are impossible. The color of each point depends on the ratio $\rho = \log(\text{MSE}[A_r]/\text{MSE}[M_r])$. When $\rho > 0$, i.e., the median is better, their color goes from light yellow to dark purple when ρ goes from 0 to 5, linearly in ρ , and is dark purple above 5. For $\rho < 0$, the color goes from light blue to dark blue when ρ goes from 0 to -0.4 . When we change r , the points remain at the same positions, but they may change color. They tend to take slightly darker color (ρ moves away from 0) when r gets smaller. With $r = 5$ (see [31]), a few points get only slightly darker than with $r = 11$.

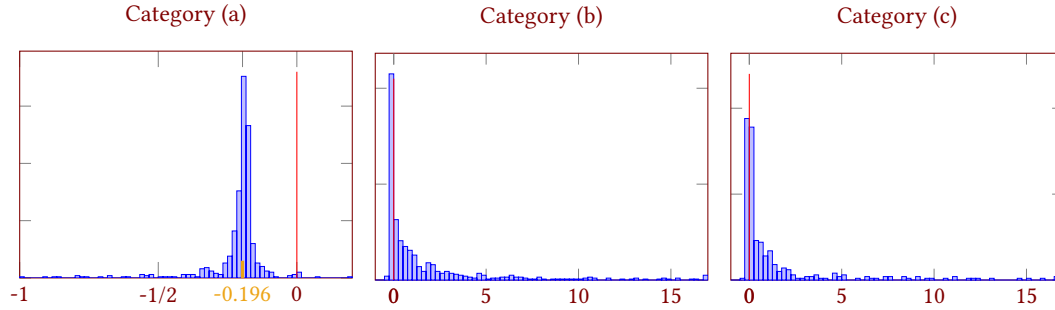


Fig. 4. Histograms of the estimated values of $\rho = \log_{10}(\text{MSE}[A_r]/\text{MSE}[M_r])$ for $r = 11$ for the three categories as in Figure 3. On the left of the vertical red line, the A_r wins, while on the right of that line, M_r wins. The orange mark at $\log_{10}(2/\pi) = -0.196$ indicates the value of ρ we would get if X had the normal distribution and $r \rightarrow \infty$, according to Example 2.7. We see that ρ is often close to that value for Category (a) even if the distribution of X obeys a CLT only in the Sob-NUS case.

the absolute skewness $|\gamma|$ for all cases, for each category of methods, (a) to (c). The color of each point depends on the ratio $\rho = \log(\text{MSE}[A_r]/\text{MSE}[M_r])$, which goes from near -1 to near 20 in our data. When $\rho > 0$, i.e., the median is better, their color goes from light yellow to dark purple when ρ goes from 0 to 5 and above, linearly in ρ . For $\rho < 0$, the color goes from light blue to dark blue when ρ goes from 0 to -0.4 . Similar plots for $r = 5$ and $r = 63$ are given in [31]. We find that with fixed parameters, the kurtosis is never very large and taking the median is never better. With random parameters, for either QMC or RQMC, the kurtosis is often quite large and then the median has a smaller MSE than the average. Recall that for Sobol' points, LMS is in the “random parameters” categories (b) and (c), but NUS is not.

Figure 4 shows histograms of the log of ratio values for each category. The largest ratios in categories (a), (b), (c) are 1.86 , 7.7×10^{22} , and 6.1×10^{19} , respectively. The very large values occur when $\text{MSE}[M_r]$ is extremely small. For comparison, for $r = 63$, those values are 78.7 , 1.1×10^{25} , and 9.0×10^{21} , respectively. One could think that for

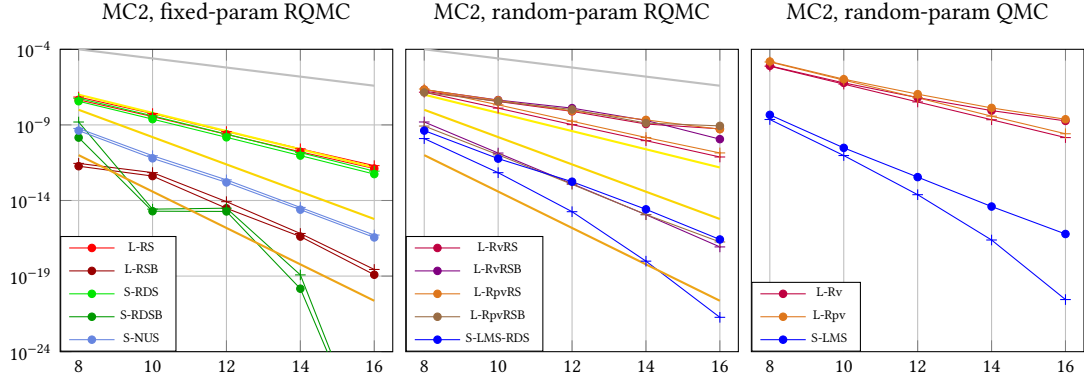


Fig. 5. $\text{MSE}[A_r]$ and $\text{MSE}[M_r]$ as functions of k , in log scale, for the MC2 integrand with $s = 4$ and $r = 11$. The legend indicates the color for each method. We abbreviate Lat to L and Sob to S. The lines with a “•” mark are for A_r and those with a “+” mark are for M_r . The three panels use the same scale to make all the plots easily comparable. The grey, yellow, dark yellow, and orange straight lines show convergence rates of n^{-1} , n^{-2} , n^{-3} , and n^{-4} , respectively.

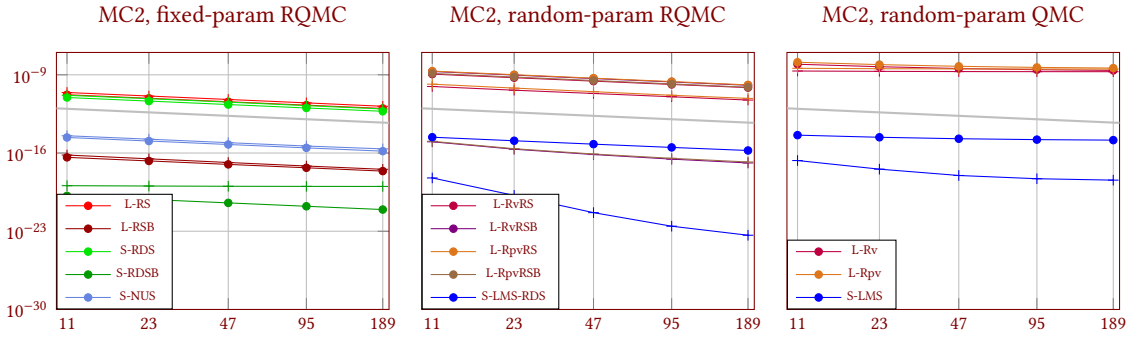


Fig. 6. $\text{MSE}[A_r]$ and $\text{MSE}[M_r]$ as functions of r , in log scale, for the MC2 integrand with $s = 4$ and $k = 14$. The lines with a “•” mark are for A_r and those with a “+” mark are for M_r . The grey line indicates a convergence rate of r^{-1} .

highly-skewed distributions, taking the median is a bad idea because it can be highly biased for the mean, but these distributions must also have a large kurtosis, and turn out to give a smaller MSE for the median, as shown in Figure 3. In Category (a), the highest yellow point is for SmoothPerB4 with Sob-NUS and the rightmost yellow point is for IndSumNormal with Sob-RDSB. The distribution of values of ρ for each category is shown in Figure 4.

5.4 Convergence of the MSE for the average vs the median

In Figure 5, we compare the convergence of $\text{MSE}[M_r]$ vs $\text{MSE}[A_r]$ in log scale, for all the methods applied to the MC2 function in $s = 4$ dimensions. The three panels correspond to the three categories of methods described in Section 5.1. For Category (a), with fixed parameters, the median M_r has a slightly larger MSE than the average A_r in all cases, and adding a baker transformation in Lat-RSB and Sob-RDSB improves the convergence rate significantly as expected, because this integrand is smooth and not already periodic. This is further examined in [36]. The convergence rates correspond to what the theory suggests. For Categories (b) and (c), with random parameters, the MSE has a better

convergence rate with M_r than with A_r in all cases, and the Sobol' points with LMS performs better than the lattice rules. Sob-RDSB is the best performer overall for this particular function. Figure 6 shows a similar convergence comparison, but for $\text{MSE}[M_r]$ and $\text{MSE}[A_r]$ as functions of r , for $r = 11, 23, 47, 95, 189$, for the MC2 function with $s = 4$ and $k = 14$. We see that at best, the MSE only decreases at the slow rate of $1/r$ when we increase r for fixed n . For the (biased) QMC methods in category (c), in most cases it is almost flat and converges to a positive constant when $r \rightarrow \infty$, not to zero. The only method for which the MSE converges a bit faster than $1/r$ is Sob-LMS-RDS combined with M_r . Similar plots in terms of both k and r for all the integrands and values of s can be found in [31].

5.5 The distributions of A_r and M_r

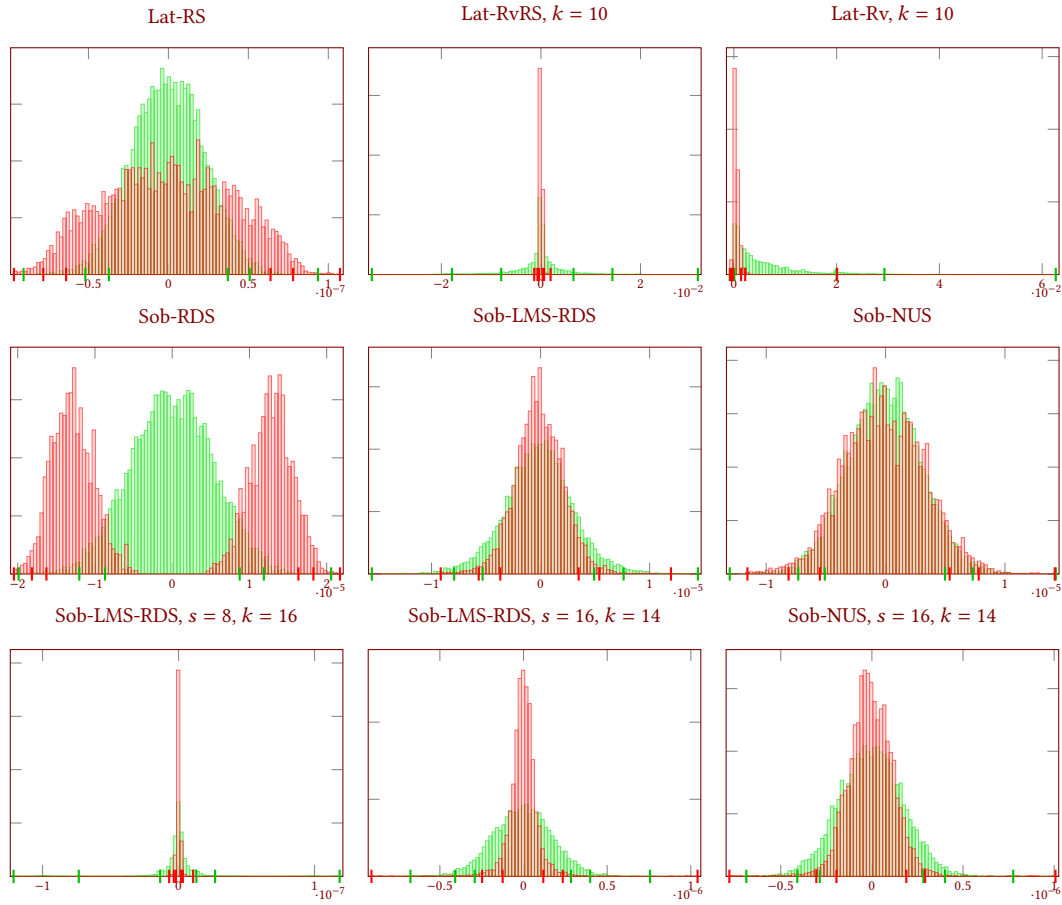


Fig. 7. Histograms of 10,000 realizations of A_r (in green) and M_r (in red), for the function SmoothPerB4 with $s = 8$ and $k = 16$ (except when k is given explicitly) in the first two rows and MC2 in the third row. The green and red marks indicate the smallest and largest observation, and the quantiles at 1%, 5%, 95%, and 99%.

Beyond comparing their MSE, we also want to look at the entire distributions of A_r and M_r , and compare their quantiles in the tails. This is particularly relevant because the available theoretical results for robust estimators, including

those in our Theorems 3.2 and 3.4, are mostly for the tail probabilities. For this, for selected values of r , for each case, we drew 10,000 observations of A_r and M_r by bootstrapping from the .dat files, and constructed the two corresponding histograms on the same plot, in green for A_r and in red for M_r . We also added green and red marks on the horizontal axis to indicate the smallest and largest observation, and the 100th and 500th smallest and largest observation, i.e., the quantiles at 1%, 5%, 95%, and 99%. The entire collection of histograms is available for display on the GitHub site.

Figure 7 shows illustrative examples for the same 9 cases as in Figures 1 and 2, except that we took $k = 10$ for the Lat-RvRS and Lat-Rv cases, to get histograms that are less degenerate. For SmoothPerB4, we can see for example that for Lat-RvRS, Lat-Rv, and Sob-LMS-RDS, the quantiles of A_r are farther away from the true mean than those of M_r , and much farther away for the lattices with random parameters. This means that occasionally, there are very bad realizations of A_r . Lat-Rv also gives highly-asymmetric histograms. That asymmetric behavior occurs in general for the biased Lat-Rv and Lat-Rpv estimators. For Lat-RS, Sob-RDS, and Sob-NUS, the quantiles of A_r are closer to the mean than those of M_r , although only slightly for Sob-NUS. For the MC2 function, the first two plots are for Sob-LMS-RDS for slightly different values of s and k . We see that M_r has better quantiles than A_r in both cases, except that in the second case, the two extreme values are worst for M_r than for A_r . This may indicate that the distribution of M_r has thicker far tails. We see a somewhat similar behavior for Sob-NUS on the right plot.

5.6 Convergence of the quantiles of A_r and M_r

Since our main theoretical results are on the quantiles of the absolute errors of A_r and M_r , it is also interesting to see how these quantiles behave in examples and how they converge as functions of k . Figure 8 shows a few examples of this, for the MC2 integrand. More examples are given in [31]. The typical trend is that for the random parameter methods, these quantiles are usually smaller and often converge to 0 faster for M_r than for A_r , as can be seen in the middle and right panels of Figure 8, but for the fixed-parameter methods there is usually not much difference.

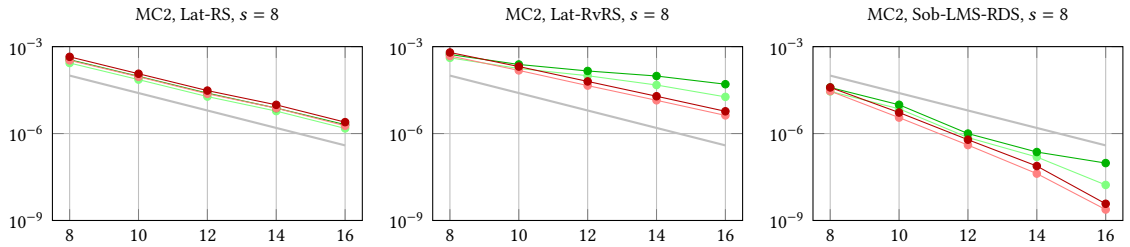


Fig. 8. The 95% and 99% quantiles of the absolute errors $|A_r - \mu|$ in green and $|M_r - \mu|$ in red, as functions of k , for the MC2 integrand with $s = 8$ and $r = 11$, with the vertical axis in log scale. The grey line indicates a convergence rate of k^{-1} .

CONCLUSION

We developed a general framework to study the median of i.i.d. estimates as an estimator of the mean and we proved properties of this estimator. We showed how to apply this framework to prove concentration inequalities for the median in settings where the estimates are replicates of a QMC or RQMC estimator with fixed or random parameters. We complemented these theoretical results with extensive numerical experiments that examine the distribution of the QMC and RQMC estimators with various integrands and compare the convergence of the MSE when taking the median vs the average. We found that taking the median can give a much smaller MSE than the average when the kurtosis of the base estimator is large, which occurs frequently for QMC or RQMC with random parameters. Using the median sometimes gives a larger MSE than using the average, but never much larger in our numerical experiments.

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